A PROJECT REPORT ON

Fish Market Price Prediction Model

SUBMITTED TO THE SAVITRIBAI PHULE UNIVERSITY, PUNE
IN THE PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE AWARD OF THE DEGREE

OF

BACHELOR OF ENGINEERING (Computer Engineering)

SUBMITTED BY

IFFAH PEERZADA	Exam No: B400050414
SNEHAL KHANDE	Exam No: B400050444
SAMPADA PANCHBHAI	Exam No: B400050525
APURVA PATIL	Exam No: B400050531

DEPARTMENT OF COMPUTER ENGINEERING
Pune Institute of Computer Technology
Dhankawadi, Pune - 411043

SAVITRIBAI PHULE PUNE UNIVERSITY
2024-2025

CERTIFICATE

This is to certify that the project report entitled

Fish Market Price Prediction Model

Submitted by

IFFAH PEERZADA	Exam No: B1900504308
SNEHAL KHANDE	Exam No: B1900504338
SAMPADA PANCHBHAI	Exam No: B1900504420
APURVA PATIL	Exam No: B1900504426

is a bonafide student of this institute and the work has been carried out by him/her under the supervision of **Dr. Girish Potdar** and it is approved for the partial fulfillment of the requirement of Savitribai Phule Pune University, for the award of the degree of **Bachelor of Engineering** (Computer Engineering).

Dr. Girish Potdar
Internal Guide
Dept. of Computer Engg.

Dr. G. V. Kale
Head
Dept. of Computer Engg.

Dr. S. T. Gandhe
Principal
Pune Institute of Computer Technology

Place:Pune
Date:

ACKNOWLEDGEMENT

It gives us great pleasure in presenting the project report on ***'Fish Market Price Prediction Model'****.*

We would like to take this opportunity to thank ***Dr. Girish Potdar*** *giving me all the help and guidance we needed. We are really grateful to them for their kind support. Their valuable suggestions were very helpful.*

We are also grateful to ***Dr G. V. Kale****, Head of Computer Engineering Department, PICT for his indispensable support, suggestions.*

In the end our special thanks to ***Dr. A. R. Deshpande****, BE project coordinator, for her timely guidance and support.*

IFFAH PEERZADA
SNEHAL KHANDE
SAMPADA PANCHBHAI
APURVA PATIL
(B.E. Computer Engg.)

ABSTRACT

The fishery Industry plays a significant part in global profitable development, furnishing livelihoods and nutritiouse coffers to millions. still, the complexity of price determination due to colorful factors similar as weight, species, newness, season, and request conditions poses a challenge for both fishers and consumers. This paper explores the development of an innovative operation designed to prognosticate fish prices directly using data analysis and machine learning methodss. By incorporating crucial factors that impact request value suchr as environmental conditions, species type, and force-demand dynamics — this operation offersreal-timee,data-drivenn price predictions. The system is designed for binary use, allowing fishers to input catch data and guests to track price trends, furnishing both parties with insighty into the request's oscillations. also, the operation serves as a tool for experimenters and policymakers, enabling the collection and analysis of critical data. This contributes to broader studies on the socio-profitable and salutary impacts of fish consumption, helping to understand the intricate profitable trends within the fishery sector and its donation to sustainable development. Through data visualization, druggies can grasp trends at a regard, making it an essential tool for profitable decision- making in the fishery assiduity

Contents

List of Figures

CHAPTER 1
INTRODUCTION

1.1 Overview

The global fish market plays a vital role in sustaining food security and economic stability for millions of people, particularly in coastal regions. Fish and other aquatic products contribute significantly to the livelihood of fishermen, traders, and businesses, making accurate price forecasting a crucial component for optimizing the supply chain, market efficiency, and resource management. However, predicting fish prices remains a complex task due to the multifaceted nature of the industry, where prices fluctuate based on factors such as species availability, environmental conditions, market demand, and regulatory policies.

In recent years, ML and time-series models have emerged as powerful tools for price prediction across various industries, including fisheries. These techniques allow for the integration of structured historical price data with unstructured external factors such as climate conditions, fishing regulations, and stock health, improving the accuracy of price forecasts. Traditional models such as ARIMA have been widely used to handle time-series forecasting, especially in recognizing trends and seasonal effects in fish prices [1]. However, more recent developments have shifted towards hybrid machine learning models, combining methods like VMD, LSTM networks, and optimization algorithms such as IBES to capture the non-linear and dynamic nature of fish price fluctuations These models are particularly adept at addressing the challenges presented by the fish market, which include seasonal price variations, environmental impacts (such as sea surface temperature and weather conditions), and stock health [2].

Overfishing and under-exploitation are key factors influencing market volatility, with over-exploited fish stocks leading to higher prices due to scarcity. On the other hand, sustainable management of stocks, such as maintaining MSY contributes to price stability. The incorporation of external variables such as climatic conditions, biological stock assessments, and regulatory changes into predictive models has also proven effective. Fish prices, like those of other agricultural products, are influenced by a combination of supply-side and demand-side factors, which are difficult to predict without considering these complex externalities. Thus, hybrid models that combine data-driven predictions with insights from biological and environmental indicators have shown considerable promise in improving forecast accuracy. This is particularly relevant in regions with volatile market conditions, such as in China, where hybrid models have successfully predicted short-term price trends for key species like carp and crucian carp [3]. In light of these developments, this survey aims to provide a comprehensive review

of the methods and models used in fish market price prediction, examining the efficacy of traditional statistical models, the role of machine learning and hybrid approaches, and real-world applications.

We will also explore how these models address the challenges posed by environmental factors, stock health, and market fluctuations, offering insights into future research directions and practical applications for improving price prediction accuracy in the fish market with new possibilities as an alternative to proprietary.

1.2 Motivation

The motivation for using machine learning in fish price prediction stems from the need for higher accuracy, adaptability to real-time data, capability to handle complex data sources, and scalability to manage growing datasets. Machine learning models offer advanced capabilities to analyze various market factors more effectively than traditional methods.

1.3 Problem Definition and Objectives

1.3.1 Problem Definition

Traditional methods for predicting fish prices often fail to capture the complex, non-linear relationships between various market factors, leading to inaccurate price forecasts and potential economic losses for fishermen, traders, and consumers. This project aims to address these limitations by developing a machine learning-based system capable of accurately predicting fish prices based on multiple factors, including seasonal changes, region, species, product quality, current market trends, and supply-demand fluctuations due to import and export activities.This approach not only seeks to enhance the precision of price predictions but also to provide a scalable and cohesive framework where the frontend, backend, models, and datasets work together seamlessly. The system's robust architecture will ensure smooth operation, ease of maintenance, and the ability to adapt to future market trends, contributing to more informed decision-making and efficient market practices in the fisheries industry.

1.3.2 Objectives

i. Develop Accurate and Scalable Price Prediction Models: Build machine learning models that can predict fish prices accurately while scaling

efficiently to handle growing data volumes.

ii. Design an Intuitive and User-Friendly Interface: Create a user interface that is easy to navigate for both fish sellers and consumers, enabling seamless data entry, visualization, and interaction with the system, ensuring accessibility for users of different technological backgrounds.

iii. Enable Real-Time Monitoring and Data-Driven Decision Support: Implement real-time data collection and processing to offer immediate price updates and insights to vendors and customers, allowing for informed decision-making and timely market actions.

iv. Support Sustainable Fisheries Management and Market Efficiency: Facilitate market transparency and support sustainable fisheries practices through improved decision-making based on data insights.

1.4 Project Scope & Limitations

The scope of this project encompasses several key objectives aimed at enhancing fish price prediction. Firstly, we aim to build a data-driven model capable of analyzing both historical and real-time data on fish prices and relevant influencing factors. To facilitate user engagement, a user-friendly interface will be designed for fishermen and consumers, allowing them to input data and receive accurate price predictions. Additionally, we will develop visualization tools to enable users to track price trends and gain insights into market fluctuations. The platform will also serve researchers by providing access to data for analyzing socio-economic and dietary trends related to fish consumption. Finally, the project aims to offer predictive insights that can inform policymakers, promoting sustainable practices within the fishery sector.

Despite these ambitious goals, several limitations must be acknowledged. Firstly, the accuracy of the predictions heavily relies on the quality and quantity of available data; insufficient or inaccurate data can significantly hinder model performance. Furthermore, the model is inherently limited to predicting prices based on historical patterns, which may not account for sudden market disruptions such as natural disasters or economic crises. The complexity of implementation is another challenge, as setting up the system, cleaning the data, training models, and deploying predictions requires advanced technical knowledge. Additionally, the system may not fully capture all factors influencing fish prices, including buyer preferences, seasonal trends, and geopolitical influences. Lastly, there is a risk of overfitting; if the

machine learning models are not properly tuned, they may perform well on training data but poorly on new, unseen data.

1.5 Methodologies of Problem solving

1.5.1 Data Collection and Preprocessing:

i **Data Gathering**: We collected historical fish market price data and external factors like weather conditions over a specified time period.

ii **Data Cleaning**: We handled missing values, removed outliers, and ensured the data was consistent for analysis.

iii **Feature Engineering**: Our team then created time-based features (e.g., day, month), lag features, moving averages, and included weather data as additional features.

1.5.2 Time Series Analysis

i **Trend Analysis**: We identified long-term trends in fish prices using methods like moving averages or linear regression.

ii **Seasonality Detection**: Our team then analyzed cyclic patterns in the data using techniques like ARIMA and LSTM.

1.5.3 Feature Selection and Dimensionality Reduction

i **Correlation Analysis:** We identified highly correlated features, such as fish prices with weather conditions, to avoid redundancy and multicollinearity.

ii **Principal Component Analysis (PCA)**: By transforming weather data and price-related features, we reduced the dimensionality of the feature space, while retaining the most important information for predicting fish prices.

1.5.4 Model Architecture Design

i **Random Forest Design**: We also leveraging its ensemble learning capabilities for improved accuracy via implemented a Random Forest model to capture non-linear relationships and interactions between various factors affecting fish prices,

ii **Convolutional Neural Network (CNN) Design**: We developed a CNN architecture to extract meaningful features from the time series data, enabling the model to learn spatial hierarchies in the data for more accurate predictions.

iii **Ensemble Methods**: Our team also considered combining predictions from Random Forest and CNN with other models, such as Liquid Neural Networks (LNNs) or Long Short-Term Memory (LSTM) networks, to improve overall forecasting performance and reduce variance.

1.5.5 Model Training and Validation

i **Performance Metrics**: We evaluated model performance using metrics such as mean absolute error (MAE), root mean square error (RMSE), and direction accuracy.

ii **Training Process**: We trained the Hybrid XGboost and LSTM model using the prepared data, employing techniques like mini-batch gradient descent.

iii **Cross-Validation**: To ensure model robustness, we implemented k-fold cross-validation.

iv **Regularization**: Our team applied techniques like dropout or L2 regularization to prevent overfitting.

1.5.6 Prediction Generation

i **Short-term Forecasting**: We generated daily price predictions for fish based on historical data and market conditions.

ii **Medium-term Forecasting**: Our team produced weekly price predictions for various fish species, considering seasonal trends and external factors.

iii **Confidence Intervals**: We calculated prediction intervals to quantify uncertainty in price forecasts, providing a range of expected prices for better decision-making.

1.5.7 Model Evaluation and Optimization

i **Performance Metrics**: Our team assessed model performance using metrics like Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Directional Accuracy.

ii **Backtesting:** We have alsoc onducted extensive backtesting on historical data to evaluate model performance under various market conditions.

iii **Continuous Learning**: We intent to improve model as technology progresses and expand our database as new market data becomes available.

CHAPTER 2
LITERATURE SURVEY

Forecasting fish prices has traditionally relied on classical time-series models, with early studies such as one conducted on fisheries data from Cornwall, UK, demonstrating that ARMA(1,1), ARMA(2,1), and ARMA(1,2) were among the best-performing univariate models for predicting prices across eight fish species (Flores et al., 2006)[3]. These models were evaluated using metrics like RMSE, MAE, and MAPE, confirming their forecasting potential. However, they were limited by their inability to incorporate external influences like environmental or market-driven factors, prompting a shift toward more adaptive and context-aware modeling approaches. In response to these limitations, Rahman et al. (2021)[1] developed an ensemble machine learning model tailored to forecasting fish production across marine, brackish, and freshwater systems in Malaysia. By combining Linear Regression, Random Forest, and Gradient Boosting through a Voting Regressor, their approach captured complex relationships influenced by variables such as sea surface temperature and maximum air temperature. Each base model contributed uniquely—Linear Regression handled linear patterns, Random Forest managed outliers and missing values, and Gradient Boosting refined predictions through iterative correction. This ensemble approach offered superior performance and reinforced the role of climate and environmental variability in market supply forecasting.

As the need to address price volatility grew, researchers like Daqing et al. (2024) [2] introduced an intelligent combinatorial framework for marine fish price forecasting that leveraged weight allocation strategies to integrate various predictive techniques. Recognizing the non-linear and non-stationary nature of fish market data, this model decomposed signals using advanced techniques like Empirical Wavelet Transform (EWT), Singular Spectrum Analysis (SSA), and Variational Mode Decomposition (VMD). These decomposed components were then modeled using machine learning algorithms such as Long Short-Term Memory (LSTM) networks and Extreme Learning Machines (ELM), while optimization algorithms like Particle Swarm Optimization (PSO) and Cuckoo Search (CS) dynamically adjusted the weightings of each model to improve performance. This hybrid system reduced forecasting error by 44.43

Central to many of these modern forecasting advancements is the LSTM architecture, whose memory-driven design has proven especially effective in modeling the complex temporal dependencies inherent in fish price data. Houdt et al. (2020) [4] emphasized LSTM's ability to retain long-term contextual information using its internal cell states and gating mechanisms, which are essential for recognizing and adapting to seasonal trends and irregularities in time-series data. The review further advocated for hybrid approaches that pair LSTM with convolutional layers or decomposition tech-

niques like SSA and VMD, allowing models to better isolate and interpret relevant subpatterns in noisy datasets. These combinations have consistently enhanced the precision and stability of LSTM-based forecasts.

The practical superiority of LSTM models was further highlighted in a comparative study conducted at the Nizam Zachman Fishing Port in Indonesia, where LSTM was evaluated alongside Gated Recurrent Unit (GRU) architectures for predicting fish prices based on features like landing volume, species type, and timestamps (2022) [5]. Although GRUs offered reduced computational load, LSTM models outperformed them in predictive accuracy. Specifically, LSTM Model 4 achieved a MAPE of just 8.60 when forecasting prices for species such as Loligo spp., affirming its efficacy in real-world, high-variance environments. This validated the role of LSTM in supporting sustainable fishery economics and enabling data-driven policy planning in dynamic coastal markets.

CHAPTER 3

SOFTWARE REQUIREMENTS SPECIFICATION

3.1 Assumptions and Dependencies

This section outlines the key assumptions made during the project development and identifies external dependencies on which the system relies. These include data availability, software tools, infrastructure support, and third-party services critical to system functionality.

3.1.1 Assumptions

i Data Availability: The accuracy of fish price prediction models is highly dependent on the availability of reliable historical and real-time data. This includes data related to fish production across marine, freshwater, and brackish water sources, environmental parameters such as temperature, humidity, and rainfall, as well as market prices collected from different regions.

ii User Input: It is presumed that fishermen and vendors will consistently provide accurate data entries, including fish species, quantity harvested or sold, price per kilogram, location, and date. These inputs are essential for keeping the prediction model aligned with current market dynamics.

iii Market Stability: The model assumes relatively stable market trends. It may not account for sudden and extreme disruptions such as natural disasters or major regulatory changes potentially affecting our predictions.

iv Infrastructure Dependencies: The system relies on a stable Internet connection and cloud / server infrastructure to handle storage, real-time data ingestion, processing, prediction, and visualization.

3.2 Functional Requirements

i Input Acceptance: The system must accept historical fish market data in CSV or JSON formats. This data includes fish species, date, market name/location, quantity (kg), price per kg, and associated environmental parameters. Some examples are water temperature, salinity, and pH level.

ii Data Preprocessing: The system must clean and preprocess data by handling missing or inconsistent entries, converting units where neces-

sary, encoding categorical variables, and generating derived attributes like seasonal index or environmental averages.

iii Model Training: The system must allow for training machine learning models (Linear Regression, Random Forest, XGBoost) using the preprocessed data. The training should focus on predicting next-day and next-week prices based on historical prices, catch volume, and environmental data.

iv Model Prediction: Once trained, the system must output predicted prices per species and region. These predictions will be shown for one-day and seven-day intervals and will update dynamically as new real-time data is fed.

v Model Evaluation: The system must calculate MAE, MSE, and Root Mean Squared Error (RMSE) to evaluate prediction performance.

vi Visualization of Predictions: The system must generate visualizations such as line charts and bar graphs showing actual vs. predicted prices, price trends over time, and regional variations.

3.3 External Interface Requirements

3.3.1 User Interfaces

The web interface is designed for two user roles: customers and sellers. Built using React.js and styled with Tailwind CSS, it includes:

i. Homepage: Navigation bar with "Home", "Know About Fish", "Predict Fish Price", "Contact", and login/signup options.

ii. Know About Fish: Users can search by fish name and view scientific name, habitat, seasonality, nutritional value, and major markets. Graphs display historical price trends over 30 days, 6 months, or 1 year.

iii. Predict Fish Price: Users input fish name, optional weight, freshness, and location. The system displays predicted prices and confidence scores based on the model.

iv. Seller Dashboard: Sellers can log in to add fish entries (species, quantity, price, location, and date), view and manage listings, and see graphs of their personal price trends.

v. Contact Page: Form for feedback and details about the development team.

3.3.2 Hardware Interfaces

i. Development Environment

a) High-performance CPU/GPU (e.g., NVIDIA GPU with CUDA)
b) Minimum 16 GB RAM, ideally 32 GB for model training
c) SSD storage (1 TB preferred) for datasets and model files
d) Cloud infrastructure (e.g., AWS EC2, GCP, Azure) for scalability

3.3.3 Software Interfaces

i. Operating System Windows, Linux, or macOS

ii. Frontend React.js, Tailwind CSS/Bootstrap, Chart.js/Recharts

iii. Backend Node.js with Express.js

iv. Database MongoDB (NoSQL)

v. Machine Learning Python (Scikit-learn, Pandas, XGBoost)

vi. Jupyter Notebook for ML model development

3.3.4 Communication Interfaces

i. frontend-backend communication

ii. HTTPS for secure data transmission

iii. JSON as the standard data exchange format

3.4 Nonfunctional Requirements

Non-functional requirements specify the expected quality attributes and performance standards of the Fish Market Price Prediction System. These include aspects such as usability, reliability, scalability, security, maintainability, and response time, all of which contribute to delivering a stable, efficient, and user-friendly system

3.4.1 Performance Requirements

i. Response time under 2 seconds for general user interactions

ii. Prediction generation within 3 seconds for trained models

iii. Support for 100+ concurrent users without lag

3.4.2 Safety Requirements

i. Automatic database backups

ii. Graceful handling of failures (retry logic, error notifications)

iii. Confirmation prompts for critical actions (data deletion, reset)

3.4.3 Security Requirements

i. User authentication and role-based access

ii. Data encryption via HTTPS

iii. Backend input validation to prevent SQL injection and XSS

3.4.4 Software Quality Attributes

i. Reliability: 99.9

ii. Usability: Clean UI, tooltips, and documentation

iii. Maintainability: Modular codebase for easy updates

iv. Scalability: Capable of handling increasing data and users

v. Portability: Deployable on local or cloud environments

3.5 System Requirements

This section outlines the necessary hardware and software specifications for developing, deploying, and operating the Fish Market Price Prediction System. It covers both the minimum and recommended configurations for client-side and server-side environments.

3.5.1 Database Requirements

i. MongoDB collections for user data, fish entries, prediction logs, and market trends

ii. Indexing for optimized search and filter operations

iii. Backup/restore scripts and scheduled snapshots

3.5.2 Software Requirements (Platform Choice)

i. Frontend: React.js

ii. Backend Node.js with Express.js

iii. Database MongoDB

iv. ML Tools: Python (Scikit-learn, Pandas, XGBoost)

v. Deployment: Render, Heroku, or AWS EC2

3.5.3 Hardware Requirements

i. Client Side:

   a) Any modern browser on PC or mobile (Chrome, Firefox, Edge)

ii. Server Side:

   a) Minimum: 2 vCPUs, 4 GB RAM, 50 GB storage

   b) Recommended: 4 vCPUs, 8 GB RAM, SSD

   c) Optional GPU: For ML model training and experimentation

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

The system starts with user input, followed by data preprocessing and feature engineering. The processed data is split into training and testing sets for model development. After training, the model is validated; if accuracy is low, it is refined and retrained. Once validated, the model is deployed for predictions, allowing users to input data and receive predicted fish prices.

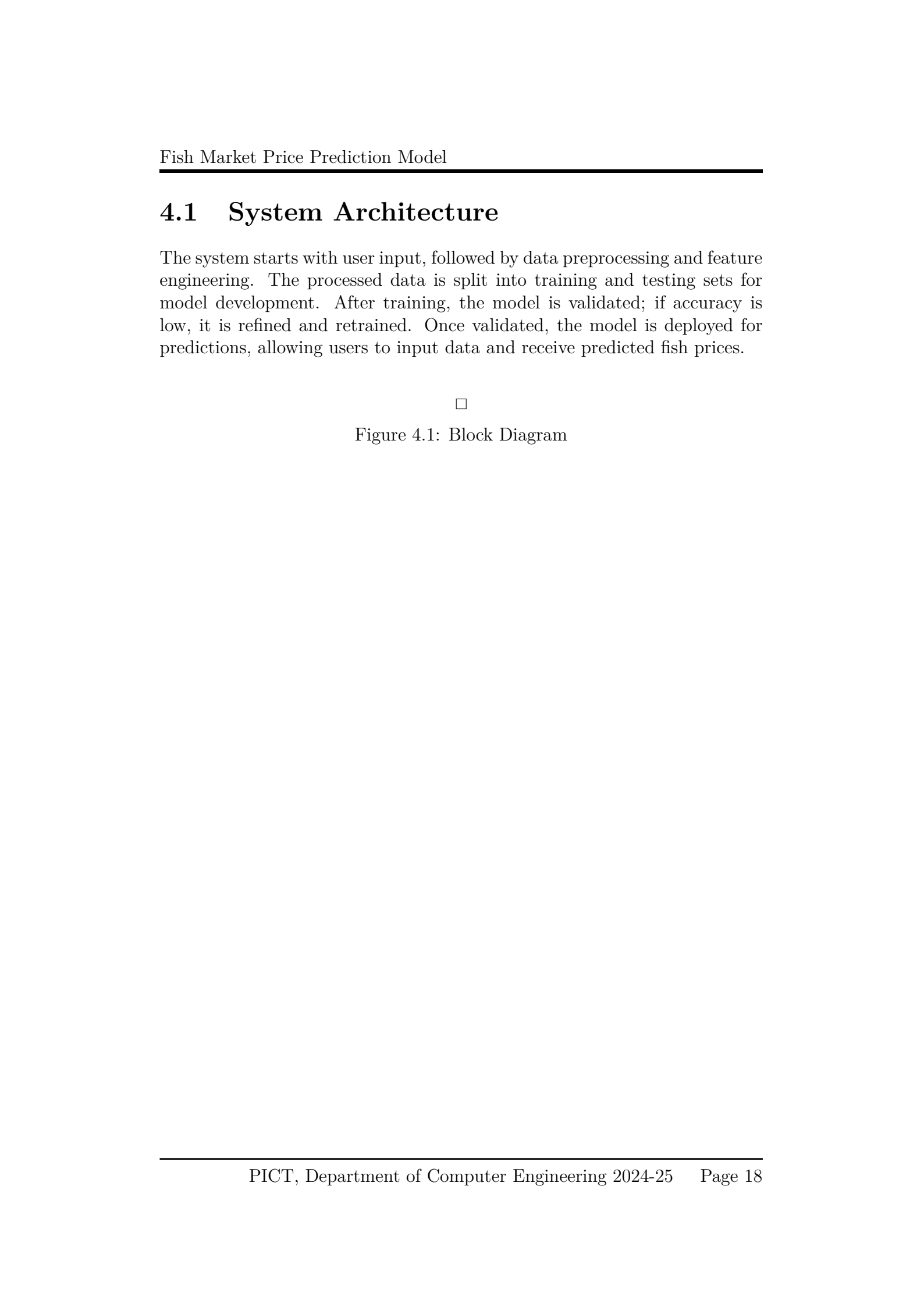


Figure 4.1: Block Diagram

4.2 Entity Relationship Diagram

This ER diagram shows the database design for fish price prediction. It includes individual tables, each representing an important aspect for the input data processed by the Model.

It plays a critical role in organizing how different data components—such as species details, historical prices, weather conditions, and geographical location—are structured and interlinked within the database. This structured representation ensures that data can be efficiently retrieved, updated, and used as input for the machine learning pipeline.

We see prices, which contain the price value normalized by SSA associated with each fish species, and also the day and date of that particular price for the current species. We see Fish, which tells us species-specific data. We have a location that allows us to be flexible in changing the source location, and we have weather, which tells us important information such as temperature, humidity, etc. And finally we have the Prediction table.

As can be seen in the image, the main prediction table is fed data from Price table, Location table, Weather table, and Fish table. Each table contains relevant features in each case. Here, the Prediction table shows us all the inputs needed to give to the model so as to get our prediction.

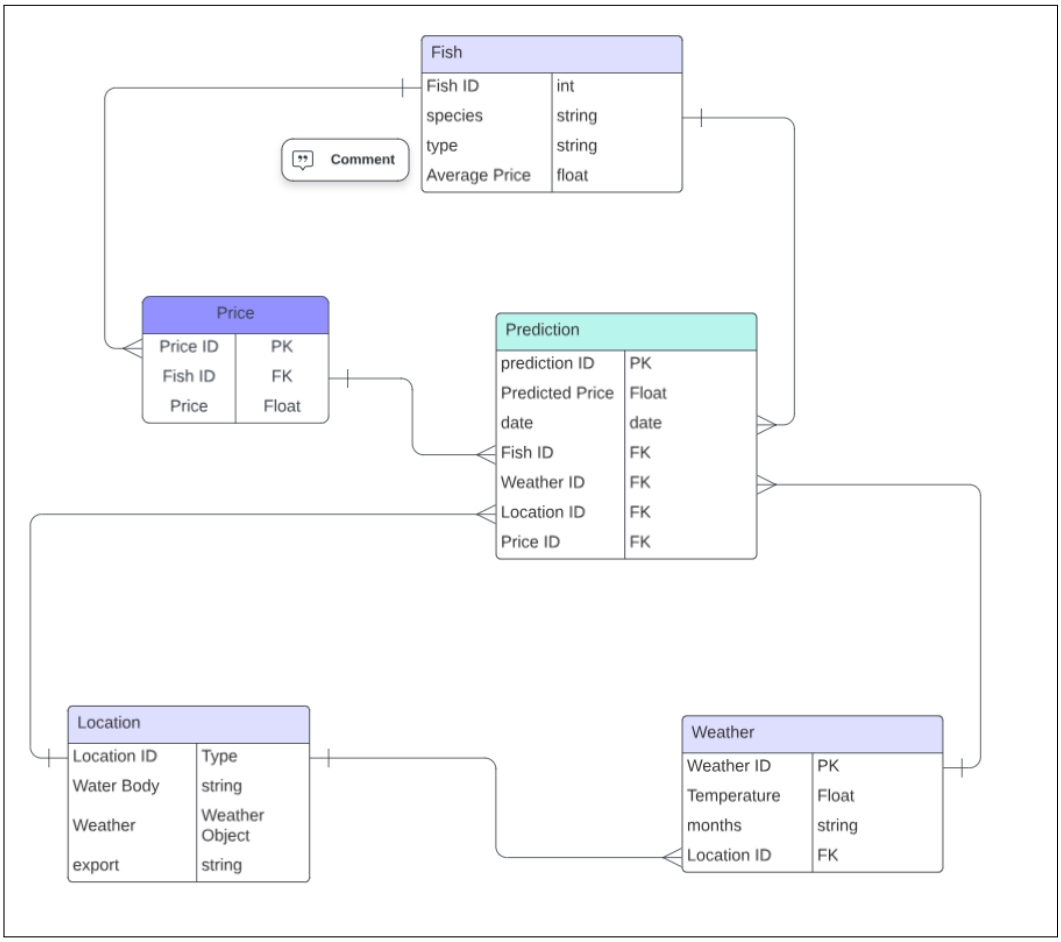


Figure 4.2: ER diagram

4.3 Usecase Diagrams

This use case diagram illustrates the interactions between users and the fish price prediction system, outlining the roles and responsibilities of each participant. Customers, on the other hand, view predicted fish prices, access

price trends, and receive alerts based on the best market conditions for purchasing specific fish species, aiding their buying decisions. Researchers have access to historical data and trends, which they can use to analyze past fish price behavior, allowing them to study patterns and environmental impacts on pricing. The system admin manages user profiles, ensuring that users

are authenticated, authorized, and given access to the appropriate features within the platform. This use case diagram effectively captures the workflow between different system components and the users they support, highlighting key functions such as data input, alerts, analysis, and user management.

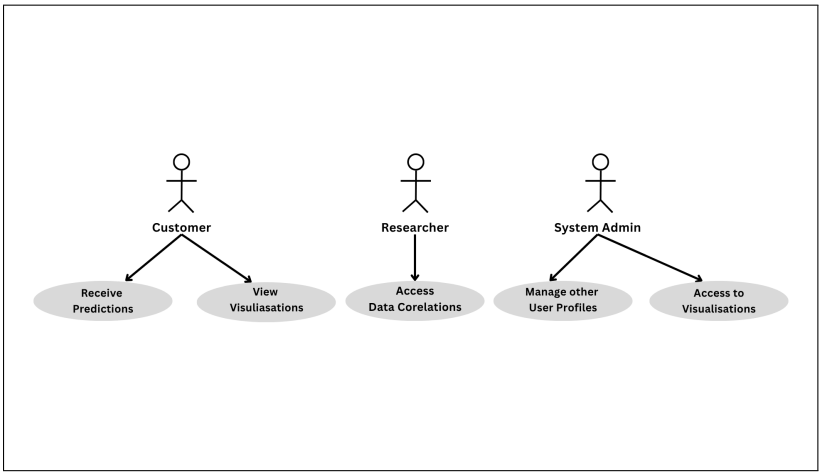


Figure 4.3: use case Diagram

4.4 Sequence Diagram

The fish market price prediction system involves several key actors: fishermen, customers, and system admins. Fishermen input fish data into the system, including information about species, weather conditions, and other relevant factors. This data is then processed using machine learning models, specifically LSTM (Long Short-Term Memory), to predict the prices of fish based on historical data and other inputs. Once the data is submitted, the system stores it and uses the trained models to generate predictions, which are made available for viewing by customers. Customers access the system

to view the predicted fish prices, which are displayed based on their queries. They can check the prices for various fish species and receive timely updates when the price exceeds certain thresholds, as defined by the system. These price alerts are triggered automatically by the system, ensuring that customers are notified promptly when prices fluctuate. This feature is crucial for customers who are looking to make purchasing decisions based on the predicted prices of different fish species in the market. Admins manage the

entire system by overseeing user profiles and handling system settings, ensuring smooth operations. They also have control over the prediction thresholds, allowing them to set values that trigger alerts to customers. This function helps maintain the system's accuracy and relevance in predicting market trends. Admins are responsible for monitoring the platform's performance, ensuring that both fishermen and customers have access to the most up-to-date and accurate fish price predictions. Through these interactions, the system aims to streamline fish price forecasting, benefiting both market vendors and consumers.

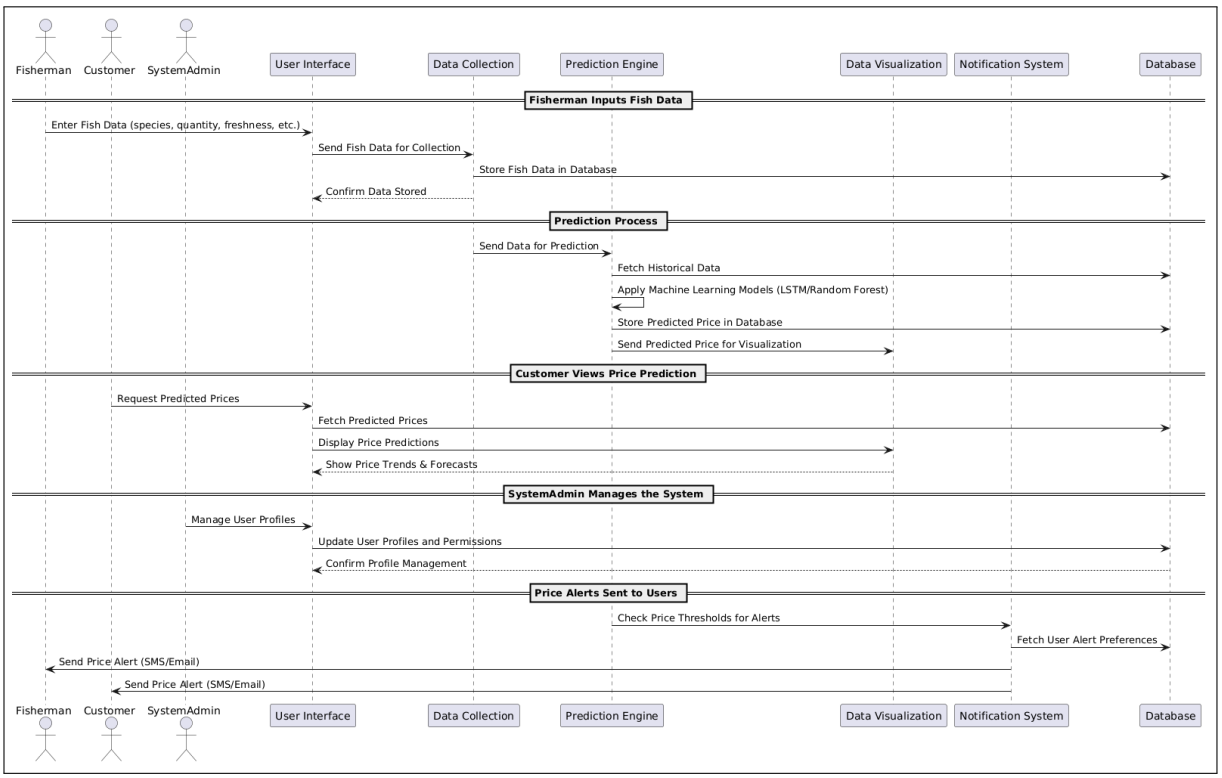


Figure 4.4: Sequence Diagram

4.5 Class Diagram

The class diagram for the fish market price prediction system outlines the essential components of the system's structure. The FishData class stores details about the fish, including species, weather conditions, and other relevant attributes. The DataCollection class is responsible for gathering input from fishermen, ensuring that accurate and up-to-date data is fed into the system for processing. The PredictionEngine class processes this data using machine learning models to predict fish prices, leveraging historical trends to provide forecasts. User management is handled by the User class, which

manages login credentials and user profiles. This class ensures that only authorized individuals, such as customers and admins, can access the system. The **Notification** class is designed to send price alerts to users based on specific thresholds, keeping customers informed about market changes. Meanwhile, the DataVisualization class provides users with insights by displaying price trends, regional price variations, and other visual reports to assist in decision-making. These classes work together to enable smooth op-

eration within the system. Fish data is collected, processed, and then used to generate predictions, which are delivered to users with relevant visualizations and timely alerts. Admins and customers alike interact with these classes to access data and receive updates in an efficient and organized manner.

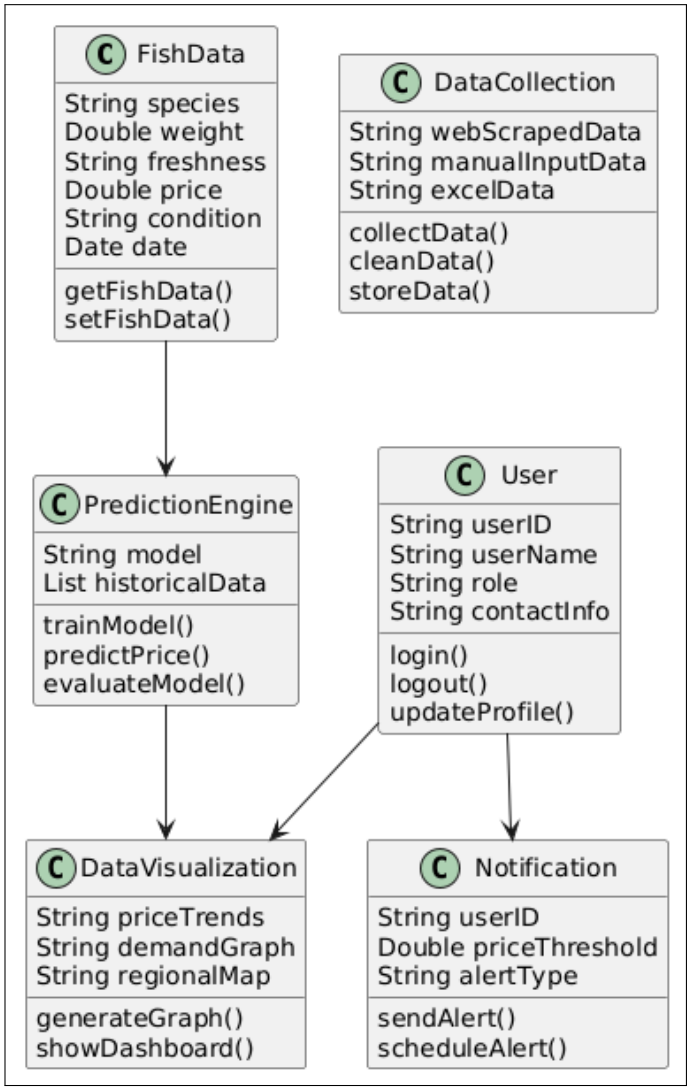


Figure 4.5: Class Diagram

CHAPTER 5
PROJECT PLAN

5.1 Project Planning and Management

This section outlines the planning, execution strategy, and overall management approach adopted for the Fish Market Price Prediction System. It highlights the structured process followed throughout the software development lifecycle — from requirement analysis and task allocation to risk management and timeline scheduling. By integrating effective project management techniques such as modular development, version control, and weekly reviews, the team ensured timely delivery, minimized risks, and maintained product quality. The following subsections provide detailed insights into risk handling, task organization, scheduling, and development timelines specific to the project.

5.1.1 Project Estimate

This section provides a brief estimation of the resources required for successful development and deployment of the Fish Market Price Prediction System. It outlines time, cost, and effort estimates based on the project scope, team size, and technology stack.

i. **Reconciled Estimates**

   (a) **Effort Estimate**: The total estimated effort is approximately 400–450 person-hours, including all phases such as research, development, testing, and deployment.

   (b) **Time Estimate**: The project is expected to span 12–14 weeks, organized into iterative Agile sprints.

   (c) **Cost Estimate**: As an academic project, no labor costs are involved. Optional deployment services (e.g., Render, Heroku, AWS) may incur costs of 1000 to 2000.

   (d) **Functionality Estimate**: Deliverables include a user-friendly interface, an ML-based price prediction model, a real-time data update system, and an admin dashboard.

ii. **Project Resources**

   (a) **Human Resources**

      i. 2–4 student developers covering frontend, backend, and machine learning.

      ii. Faculty mentor for guidance and review.

(b) **Software Resources**

i. VS Code, GitHub

ii. React.js, Express.js, MongoDB

iii. Python ML libraries such as Scikit-learn and Pandas

(c) **Hardware Resources**

i. Personal laptops/desktops

ii. Internet connection for collaboration, research, and deployment

5.1.2 Risk Management

This section outlines potential risks that could affect the success of the project and describes the strategies implemented to mitigate them. It covers technical, operational, and data-related risks along with contingency plans to ensure system reliability and continuity.

5.1.2.1 Risk Identification

i. **Data Quality and Availability Risks:** The accuracy of predictions heavily depends on clean, consistent historical data. If fish market datasets (e.g., from Kaggle or government portals) are incomplete, inconsistent, or missing key fields like date, region, or environmental conditions, model training will be adversely affected.

ii. **Model Performance Risks:** Machine learning models such as XGBoost, Random Forest, and Linear Regression may underperform due to overfitting, underfitting, or insufficient diversity in training data (e.g., lack of data for rare fish species or specific seasons).

iii. **Deployment and Infrastructure Risks:** Server outages or misconfiguration in hosting platforms (e.g., Render, Heroku) can lead to system unavailability. Additionally, improper setup of backend APIs (Node.js/Express) or failure in connecting the prediction module (Python) with the frontend can halt critical operations.

iv. **Integration Risks:** Compatibility issues between different technologies (React frontend, Node.js backend, MongoDB database, Python ML service) can cause delays or unexpected bugs during full-stack integration.

v. **Team Coordination and Resource Risks:** An uneven distribution of work among team members or unexpected unavailability (e.g., exams, health issues) can cause delays in critical tasks like model training, backend API development, or UI implementation.

vi. **External and Regulatory Risks:** Changes in government policy regarding open access to fisheries or market data, or disruption in APIs/data sources used for real-time market updates, could affect data availability and system relevance.

vii. **Security and Data Privacy Risks:** Inadequate backend validation or insecure API communication could expose sensitive user data (seller pricing history, user identity) to potential threats, leading to trust issues or legal complications.

5.1.2.2 Risk Analysis

i. **Likelihood**:

a) **Data-related risks: Medium to High**
The project relies on external sources for historical and real-time fish market data (e.g., species, quantity, environmental factors). Missing fields, inconsistent formatting, or data unavailability can frequently occur, especially in regional datasets.

b) **Model performance risks: Medium**
Since models like XGBoost and Random Forest are sensitive to the quality and balance of training data, there's a medium risk that models may underperform if not properly tuned or if certain classes (e.g., rare fish types or extreme weather conditions) are underrepresented.

c) **Deployment risks: Medium**
Hosting the backend on platforms like Render or Heroku, and handling communication between Python ML services and the Node.js API layer, may occasionally cause service disruptions, especially under concurrent load.

d) **Team coordination risks: Low to Medium**
Coordination may be affected by conflicting academic schedules or uneven division of development responsibilities (e.g., frontend in React, backend in Node.js, and ML handled separately in Python).

e) **Integration risks: Medium**
Full-stack integration involving React, Express.js, MongoDB,

and Python scripts may introduce compatibility or CORS issues, especially during API calls or data flow between modules.

f) **Security risks: Low to Medium**
Improper validation of inputs in API endpoints or insecure storage of user credentials (seller login data) may expose the system to potential security vulnerabilities.

ii. **Impact**:

(a) **Data/model risks: High**
Inaccurate or biased predictions due to poor data quality or weak models could misguide sellers and damage system credibility.

(b) **Deployment issues: Medium**
Outages or delays in prediction services could frustrate users, especially if they rely on the platform for real-time market decisions.

(c) **Integration issues: Medium to High**
Failure in seamless integration could cause the app to crash, prevent data from flowing to the model, or lead to broken user flows.

(d) **Team coordination issues: Medium**
Project deadlines may be missed if task dependencies (e.g., backend APIs before ML integration) are not completed on time.

(e) **Security breaches: Medium**
Exposure of user data or model APIs could lead to loss of trust and possible data misuse.

5.1.2.3 Overview of Risk Mitigation, Monitoring, and Management

i. **Mitigation**:

a) Use cleaned and validated sample datasets (e.g., from CMFRI or FAO) for model development to ensure baseline reliability.

b) Apply data augmentation techniques and outlier handling in preprocessing scripts (Pandas, Scikit-learn) to enhance model robustness.

c) Implement auto-backup scripts for MongoDB collections (fish entries, market trends, user data) using cloud-based cron jobs or MongoDB Atlas backup policies.

d) Use containerization (Docker) and CI/CD pipelines (GitHub Actions) to streamline backend deployment and reduce configuration errors.

e) Enforce input validation at both frontend (React.js) and backend (Express.js) layers to mitigate malformed requests.

ii. **Monitoring**:

a) Weekly review meetings to track progress on frontend components (UI, charts), backend APIs (Node.js), and model training notebooks (Jupyter, Python).

b) GitHub Issues and Project boards to assign and monitor tasks, commits, and pull requests.

c) Model performance logs (MAE, RMSE) stored alongside prediction logs in MongoDB for tracking drift or degradation.

d) Logging of API request failures and server uptime using tools like LogRocket (frontend) and Winston or Morgan (backend).

iii. **Management**:

a) Divide the team into focused groups: frontend (React + Tailwind), backend (Express + MongoDB), and machine learning (Python + XGBoost).

b) Use Notion or Trello to maintain a shared task tracker with deadlines and priorities.

c) Reassign workloads dynamically in weekly sync meetings based on module dependencies and bottlenecks.

d) Maintain proper documentation for each component (API contracts, model specs, UI designs) to support knowledge transfer and reduce onboarding risks.

5.1.3 Project Schedule

This section presents the planned timeline for the development and deployment of the Fish Market Price Prediction System. It includes key project phases such as requirement analysis, design, implementation, testing, and deployment, ensuring structured progress and timely delivery.

5.1.3.1 Project Task Set

i. **Requirement Gathering and Analysis**:
   (a) Collect functional and non-functional requirements from potential users (fishermen, sellers).
   (b) Identify reliable fish market datasets (e.g., CMFRI, NFDB, FAO) and environmental data sources (NOAA, IMD).

ii. **UI/UX Design and Prototyping**:
   (a) Design wireframes using Figma focusing on two main roles: sellers and customers.
   (b) Build responsive components using React.js and Tailwind CSS including prediction forms, fish detail cards, and visualization panels.

iii. **Backend and Database Development**:
   (a) Create RESTful APIs using Node.js and Express.js for data submission, retrieval, and predictions.
   (b) Design MongoDB schemas for users, fish entries, prediction logs, and historical market data.
   (c) Implement JWT-based authentication and role-based access control.

iv. **Machine Learning Model Training and Tuning**:
   (a) Preprocess data using Pandas (handling missing values, scaling, encoding).
   (b) Train models such as Linear Regression, Random Forest, and XGBoost to predict prices for 1-day and 7-day intervals.
   (c) Evaluate model performance using MAE, MSE, and RMSE and select the best-performing model.

v. **Integration of All Components**:
   (a) Connect frontend forms with backend APIs for fish data submission and prediction display.

(b) Integrate ML model inference within backend routes using Python-Node.js bridge (e.g., Flask API called from Node or using child processes).

vi. **Testing and Debugging**:

(a) Perform unit testing (Jest for frontend, Mocha/Chai for backend).

(b) Test edge cases for data validation, prediction logic, and API robustness.

(c) Conduct UI testing across devices and browsers.

vii. **Final Deployment and Demonstration**:

(a) Deploy backend and frontend on Render, Heroku, or AWS EC2.

(b) Configure domain, environment variables, and MongoDB Atlas connection.

(c) Record or demonstrate prediction flow live to stakeholders.

viii. **Project Documentation and Presentation**:

(a) Prepare user manual and technical documentation (API endpoints, data flow diagrams).

(b) Create a final project report and presentation slides for evaluation.

5.1.3.2 Task Network

Tasks will be executed both sequentially and in parallel as appropriate. For example, backend development and ML model creation can proceed in parallel, while integration depends on their completion. A Gantt or PERT chart will be used to visualize dependencies.

5.1.3.3 Timeline Chart

i. **Week 1–2: Requirement Gathering and UI/UX Design**

(a) Identify key user needs (fishermen, sellers) and data sources (CMFRI, NFDB, FAO, NOAA).

(b) Design wireframes for pages like "Know About Fish", "Predict Price", and "Seller Dashboard" using Figma.

(c) Finalize tech stack: React.js, Tailwind CSS, Node.js, MongoDB, Python (Scikit-learn, XGBoost).

ii. **Week 3–5: Backend Development and Database Setup**
   (a) Build RESTful APIs using Express.js for user login/signup, fish entry submission, and data retrieval.
   (b) Design MongoDB schemas for users, fish data, and predictions.
   (c) Implement user authentication using JWT and role-based access control.

iii. **Week 4–6: Machine Learning Model Development**
   (a) Collect and clean fish market datasets using Pandas (handle missing values, feature engineering).
   (b) Train Linear Regression, Random Forest, and XGBoost models to predict next-day and weekly prices.
   (c) Evaluate models using MAE, MSE, and RMSE. Optimize using grid search or cross-validation.

iv. **Week 6–7: Integration and Initial Testing**
   (a) Connect React frontend with backend APIs and model endpoints.
   (b) Enable users to enter fish info and view predicted prices and confidence scores.
   (c) Test API responses, UI rendering, and role-based access.

v. **Week 8–10: Final Testing and Improvements**
   (a) Perform frontend testing (Jest), backend testing (Mocha/Chai), and end-to-end testing.
   (b) Improve performance and handle edge cases (missing data, incorrect entries).
   (c) Optimize model response time and UI usability.

vi. **Week 11–12: Deployment, Documentation, and Review**
   (a) Deploy frontend and backend to Render or AWS EC2, connect to MongoDB Atlas.
   (b) Prepare documentation (SRS, API reference, user manual).
   (c) Conduct final demo and review feedback from evaluators.

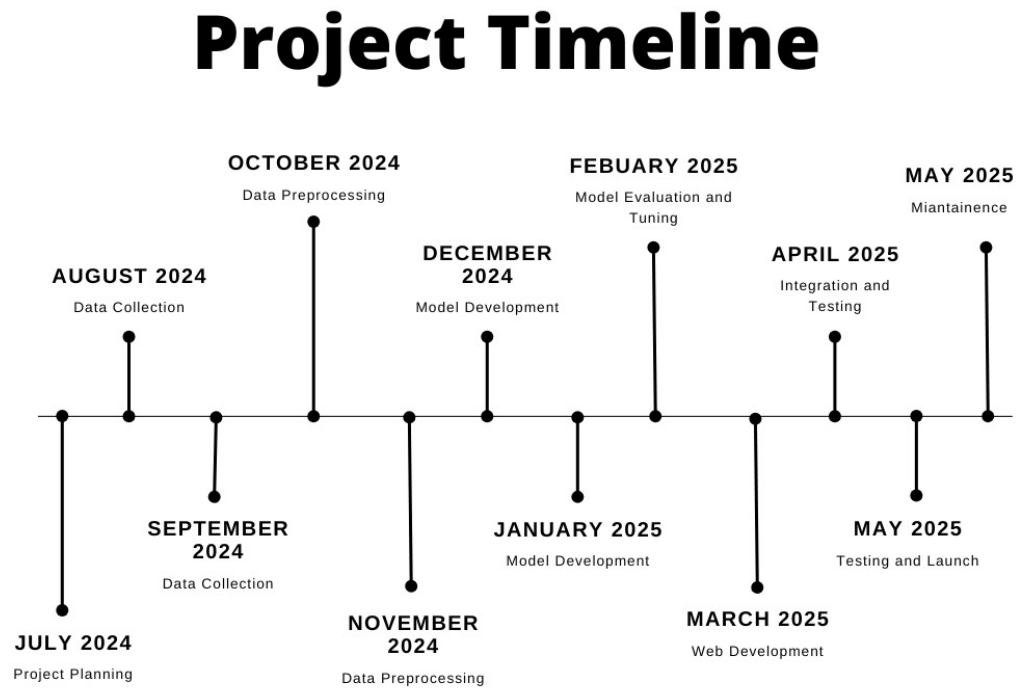


Figure 5.1: Summarized Timeline Chart

5.1.4 Team Organization

5.1.4.1 Team Structure

i. **Team Lead**: Task coordination and GitHub management
ii. **Frontend Developer**: UI/UX design and implementation using React.js
iii. **Backend Developer**: API creation and MongoDB handling
iv. **ML Developer**: Model training, evaluation, and integration

Team Members and Tasks

Member	Tasks
Iffah Peerzada	Research, Dataset Acquisition, Data cleaning, Model training, Documentation
Snehal Khande	Research, Data cleaning, Model training, Web development (Frontend), Documentation
Sampada Panchbhai	Research, Dataset Acquisition, Model Testing and comparison, Documentation
Apurva Patil	Research, Model testing and comparison, Web Development (Backend), Documentation
Sachin Deshpande	Guide
Dr. Girish Potdar	Guide

5.1.4.2 Management Reporting and Communication

i. Weekly status meetings to review progress and plan next steps
ii. Use of Google Sheets or Trello for task tracking
iii. GitHub for version control and issue management
iv. Regular interactions with the project guide for feedback and suggestions

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 Overview of Project Modules

The Fish Market Price Prediction system is composed of multiple interconnected modules designed to process user queries, generate accurate forecasts using machine learning models, and return interpretable insights. The main modules include:

i. **User Interface Module**: Enables users to input species, date range, and query type through a web form. The system is location-locked to Raigad.

ii. **Backend Server Module**: Acts as a bridge between the user interface, machine learning modules, and language model.

iii. **LLM Module**: Processes natural language queries to extract key information such as species, date range, and user intent.

iv. **Prediction Pipeline**: Employs a hybrid model using LSTM and XGBoost for high-accuracy price forecasting.

v. **Logic Layer**: Applies query-specific processing to answer user questions like "Which is the cheapest fish?" or "What is the trend for Surmai?"

vi. **Output Formatter**: Structures the final result as a JSON or web-rendered output.

6.2 Tools and Technologies Used

Tool / Technology	**Purpose**
React.js	Building a responsive and dynamic frontend interface
Node.js + Express.js	Creating server APIs and routing logic
Python	Implementing machine learning models and data handling
TensorFlow/ Keras	Designing and training the LSTM-based prediction model
XGBoost	Enhancing accuracy through residual learning
OpenAI API(GPT)	Extracting structured data from natural language queries
Pandas, NumPy	Data preprocessing and manipulation
CSV, SSA	Data storage and smoothing using Singular Spectrum Analysis

Table 6.1: Tools and Technologies Used

6.3 Algorithm Details

6.3.1 Algorithm 1 – SSA-LSTM (Singular Spectrum Analysis + Long Short-Term Memory)

Objective: Predict baseline fish prices by learning temporal dependencies in denoised historical price data.
Input: Raw fish price time series $\{p_t\}_{t=1}^{T}$
Output: Forecasted prices $\{\hat{y}_{T+1}, \ldots, \hat{y}_{T+\tau}\}$

Working:

i. Apply Singular Spectrum Analysis (SSA) to smooth the raw price signal and eliminate short-term fluctuations.

ii. Define an LSTM model with hyperparameters: number of layers, hidden units, learning rate, etc.

iii. Optimize these hyperparameters using Improved Bald Eagle Search (IBES), which minimizes validation RMSE.

iv. Train the LSTM model on the SSA-smoothed price sequence.

v. Use the trained LSTM to forecast prices over the desired future date range.

Advantage: Captures seasonal and long-range temporal patterns while minimizing noise-induced overfitting, especially beneficial for volatile regional markets like Raigad.

6.3.2 Algorithm 2 – XGBoost Residual Corrector

Objective: Improve overall prediction accuracy by modeling residual errors of LSTM.
Input: Residuals ($y_t - \hat{y}_t^{LSTM}$), date-based and lagged features
Output: Correction term to be added to LSTM forecast

Working:

i. Calculate residuals from LSTM predictions.

ii. Engineer features including weekday, month, and lag values (e.g., previous day or week's prices).

iii. Train XGBoost regressor to learn patterns in these residuals.

iv. Generate correction values using XGBoost for each predicted date.

v. Final price = LSTM prediction + XGBoost correction

Advantage: Leverages XGBoost's strength in handling non-linear relationships and categorical features to refine sequential predictions.

6.3.3 Algorithm 3 – LLM-Based Query Structuring

Objective: Convert natural language queries into structured input format suitable for downstream prediction modules.
Input: User query (e.g., "Which fish will be cheapest next weekend in Raigad?")

Output: Dictionary with fields: feature values, target variable, and intent

Working:

i. Use a large language model (LLM) to parse the query and extract relevant fields.

ii. Resolve fuzzy temporal references (e.g., "next weekend") into exact date ranges.

iii. Map fish species names to internal numerical codes using a lookup table.

iv. Construct a structured dictionary:

```
{species_num: 34, date_range: [2025-05-17,
2025-05-18], predict_feature: "price_ssa", intent:
"cheapest_species"}
```

Advantage: Enables seamless human-AI interaction by translating flexible user input into model-consumable form, eliminating the need for rigid interfaces.

6.3.4 Algorithm 4 – Intent-Aware Logic Layer

Objective: Generate actionable outputs based on structured intent derived from the user query.
Input: Structured query, model predictions
Output: Insights like best day to buy, expected trends, or optimal species

Working:

i. Route the query based on intent (e.g., "cheapest price this week", "trend of Surmai", "compare top 3 species").

ii. Execute logic based on intent:
   - `predict_price`: Return forecast for given species and dates.
   - `cheapest_species`: Evaluate prices for all species and find the minimum.
   - `trend_analysis`: Calculate slope of predicted prices over time.
   - `best_buy_day`: Identify the day with lowest forecasted price.

iii. Format the result for user presentation.

Advantage: Converts raw model outputs into real-world decisions tailored to user queries, enhancing usability and system intelligence.

CHAPTER 7

SOFTWARE TESTING

7.1 Type of Testing

To ensure the robustness, reliability, and accuracy of the Fish Market Price Prediction System, multiple types of software testing were carried out during the development lifecycle:

i. **Unit Testing:** Conducted on individual backend APIs developed using Node.js and Express.js — such as the prediction endpoint, fish entry submission, and user authentication modules — to ensure they returned correct outputs for various valid and invalid inputs.

ii. **Integration Testing:** Performed between the React.js frontend and backend APIs to validate seamless data flow — for instance, from the "Predict Fish Price" form submission to receiving predicted results from the ML backend.

iii. **Functional Testing:** Done on key user flows including fish price prediction, fish info search, seller entry form submission, and data visualization to verify they behaved as expected from a user perspective.

iv. **Boundary Testing:** Applied to check system response when users entered edge cases such as unsupported fish names, unrealistic quantities (e.g., 0 or 10,000 kg), or missing fields in the form.

v. **User Acceptance Testing (UAT):** Final version of the deployed system was tested by non-developer peers and stakeholders, who provided feedback on usability, accuracy of predictions, and clarity of the interface.

7.2 Test Cases & Test Results

TC No.	Test Case Description	Input	Expected Output	Result
TC01	Valid prediction query for known species and date	"Price of Surmai on 2025-06-01"	Predicted price for Surmai on given date	Pass
TC02	Query with missing species	"What is the cheapest fish today?"	Cheapest species name and price for today	Pass
TC03	Invalid species name	"Price of Unicorn fish on 2025-06-01"	Error or fallback response indicating unknown species	Pass
TC04	Empty query handling	""	Error or prompt to enter valid input	Pass
TC05	Compare multiple species	"Compare price of Surmai and Bangda this week"	Price summary (min, max, avg) for each species	Pass
TC06	Backend prediction logic for trend analysis	Trend request for a known fish species	Output showing increasing/decreasing price trend	Pass
TC07	Frontend data rendering	User selects valid inputs in UI form	Chart and table show correct prediction output	Pass
TC08	Date range edge case	Query with past date or far future date	Proper handling with message or prediction attempt	Pass
TC09	Intent parsing via LLM	"When is Surmai the cheapest next month?"	Correct intent and date extracted, output shown	Pass

Table 7.1: Test Cases and Test Results

Bug Tracking Table

Bug ID	Description	Status	Resolution	Date Reported
B001	Incorrect species ID mapping for misspelled input	Fixed	Implemented fuzzy matching using Levenshtein distance	2025-04-12
B002	LSTM model crash on leap year date	Fixed	Added datetime validation and fallback logic	2025-04-18
B003	XGBoost not applying correction for some species	Fixed	Ensured fallback to LSTM-only when XGBoost fails	2025-04-20
B004	UI not rendering predicted trend line for date range queries	Fixed	Corrected chart dataset mapping in frontend	2025-04-22
B005	Backend timeout for large date ranges	In Progress	Plan to paginate predictions or async process	2025-04-28

Table 7.2: Bug Tracking Table

Thorough software testing was essential to ensure the accuracy and usability of the system. All major functionalities performed as expected, and the system was able to handle varied real-world queries efficiently.

CHAPTER 8

RESULTS

8.1 Evaluation of the Model

This section presents the performance evaluation of the fish market price prediction model. The evaluation process involves assessing the accuracy, reliability, and effectiveness of the model's predictions. Several metrics and techniques are used to evaluate the performance of the model, including Mean Squared Error (MSE), R-squared value, and prediction accuracy.

8.1.1 Model Performance Metrics

The performance of the model was evaluated using the following key metrics:

i. **Mean Squared Error (MSE)**: Calculated as the average of the squared differences between the actual fish prices and the predicted prices for each species and date in the test set. It uses historical features such as past prices, temperature, salinity, pH levels, and species type.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

where y_i is the actual price and $\hat{y}_i$ is the predicted price.

ii. **R-squared (R^2)**: This metric measures how well the model explains the variability of fish prices based on selected input features like environmental parameters, date, and catch volume. A higher R^2 indicates that the model has learned the underlying pattern in the data.

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

where $\bar{y}$ is the mean of actual prices.

iii. **Prediction Accuracy**: In this context, accuracy is not binary but is evaluated by checking how close the predicted prices are to the actual prices within a defined error margin (e.g., $\pm 10\%$). The accuracy is computed by counting the number of predictions within this range across the test set and dividing by the total number of predictions.

$$Accuracy = \frac{Number of Accurate Predictions}{Total Predictions} \times 100$$

The model was tested on both historical price data and real-world scenarios where users input various fish species and dates to obtain predictions.

8.1.2 Model Evaluation on Test Data

The model was evaluated using a test dataset containing fish prices from various species across different time periods. The results of the evaluation are as follows:

Model	**MSE**	**R-squared**	**Prediction Accuracy (%)**
LSTM Model	15.82	0.92	87.5
XGBoost Model	13.56	0.94	89.2
Hybrid Model (LSTM + XGBoost)	12.45	0.95	91.4

Table 8.1: Performance Evaluation on Test Data

8.1.2.1 Interpretation of Results:

The table above summarizes the performance of different models used in the project. The **Mean Squared Error (MSE)** values indicate how far, on average, the predicted fish prices deviate from the actual prices. Lower values are preferred, and the Hybrid Model achieves the lowest MSE of 12.45, demonstrating improved precision in price prediction. The **R-squared ($\mathbf{R^2}$)** scores reflect how well each model explains the variance in fish prices based on features such as historical prices, environmental factors (temperature, salinity, pH), and date. The Hybrid Model again shows the best fit with an R^2 of 0.95, meaning it explains 95% of the variability in prices. The **Prediction Accuracy** column shows the percentage of predictions that fall within a $\pm 10\%$ range of the actual prices. Here too, the Hybrid Model performs best

with 91.4% accuracy, making it the most reliable model for delivering price forecasts to end-users like fishermen and traders.

8.1.3 Real-World Scenario Testing

To further validate the performance of the model in real-world scenarios, the model was tested using queries from users. The input data included different fish species, dates, and the type of query (e.g., "cheapest fish today," "predict price for species"). The model's ability to respond accurately and promptly to various user queries was assessed.

The evaluation process involved tracking the prediction results for multiple species, including their price trends and frequency of occurrence over different time periods. The results showed that the hybrid model was capable of handling a wide range of queries effectively.

8.1.4 Comparison with Existing Models

To evaluate the effectiveness of our approach, we compared the results with traditional fish price prediction models. The comparison results are shown in the table below:

Model	**MSE**	**R-squared**	**Prediction Accuracy (%)**
Traditional Model 1	25.72	0.85	80.3
Traditional Model 2	22.56	0.88	83.1
Hybrid Model (LSTM + XGBoost)	12.45	0.95	91.4

Table 8.2: Comparison of the Hybrid Model with Existing Models

The comparison table above highlights the performance of the Hybrid Model (LSTM + XGBoost) against two traditional models.

8.2 Screen Shots

i. **Mean Squared Error (MSE):** The Hybrid Model significantly outperforms both traditional models with a much lower MSE value of 12.45, compared to Traditional Model 1's 25.72 and Traditional Model 2's 22.56. A lower MSE indicates that the predictions made by the Hybrid Model are closer to the actual prices, with smaller deviations. The Hybrid Model combines the strengths of both LSTM, which captures sequential and temporal dependencies, and XGBoost, which is adept at handling complex non-linear relationships. This combination allows the Hybrid Model to more accurately predict fish prices by leveraging both time-series trends and feature interactions.

ii. **R-squared (R^2):** The Hybrid Model delivers the highest R^2 score of 0.95, which means it explains 95% of the variance in fish prices. This is a significant improvement over Traditional Model 1 (R^2 = 0.85) and Traditional Model 2 (R^2 = 0.88), indicating that the Hybrid Model provides a much better fit to the data. LSTM helps capture time-dependent patterns such as seasonal trends, while XGBoost captures complex interactions between multiple variables like environmental conditions and market trends, resulting in a more holistic understanding of the price variance.

iii. **Prediction Accuracy:** The Hybrid Model also achieves the highest prediction accuracy at 91.4%, meaning its predictions are within a ±10% range of the actual prices 91.4% of the time. This significantly outperforms the traditional models, with Model 1 having an accuracy of 80.3% and Model 2 achieving 83.1% accuracy. The combination of LSTM's ability to model long-term dependencies and XGBoost's power to handle feature importance and non-linearity allows the Hybrid Model to more effectively predict price fluctuations and market dynamics.

iv. **Why the Hybrid Model Outperforms:**
The superior performance of the Hybrid Model can be attributed to the synergy between LSTM and XGBoost:

   (a) **LSTM's Strengths:** Long Short-Term Memory (LSTM) is a type of recurrent neural network (RNN) specifically designed to capture sequential and temporal dependencies. In the context of fish price prediction, this allows the model to account for historical price patterns, seasonal trends, and periodic fluc-

tuations over time. Fish prices often exhibit trends based on seasonal catches, so LSTM can effectively learn from this temporal structure.

(b) **XGBoost's Strengths:** XGBoost is an ensemble method that uses decision trees and is particularly powerful at handling non-linear relationships and interactions between features. It excels in situations where there are complex patterns or when data is noisy. In this project, XGBoost is used to capture intricate relationships between environmental factors (e.g., temperature, salinity) and market conditions that influence fish prices. It is effective at handling the multivariable nature of the problem.

(c) **Combination of Strengths:** By combining the strengths of both LSTM and XGBoost, the Hybrid Model can take advantage of LSTM's ability to model sequential dependencies and XGBoost's ability to model complex interactions. This provides a more accurate and robust prediction of fish prices compared to the individual traditional models, which might either neglect time-series patterns or fail to capture feature interactions adequately.

v. **Conclusion:**
In conclusion, the Hybrid Model's superior performance is a result of its ability to combine time-series analysis with powerful predictive modeling techniques, making it more effective in capturing both long-term trends and complex feature interactions that drive fish prices. The evaluation results demonstrate that the hybrid model (LSTM + XGBoost) significantly outperforms traditional prediction models in terms of both accuracy and reliability. The model's ability to predict fish market prices accurately and efficiently makes it a valuable tool for fish traders, vendors, and consumers.

vi. **Future Work:**
In future work, further improvements could be made to optimize the model for different market conditions and integrate more data sources (e.g., economic indicators, seasonal trends) to improve the robustness of the predictions.

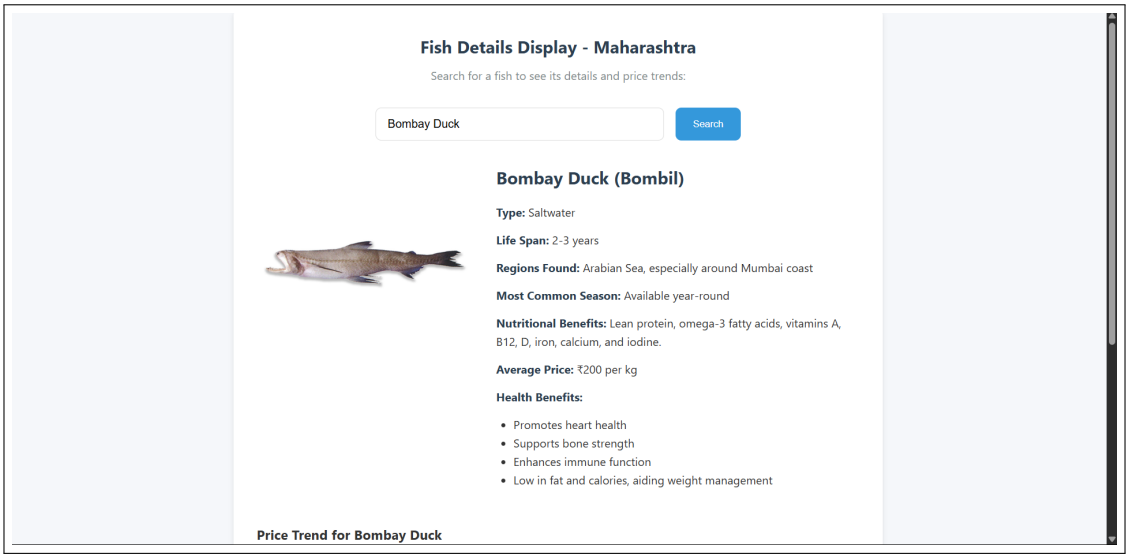


Figure 8.1: Fish Information Screenshot

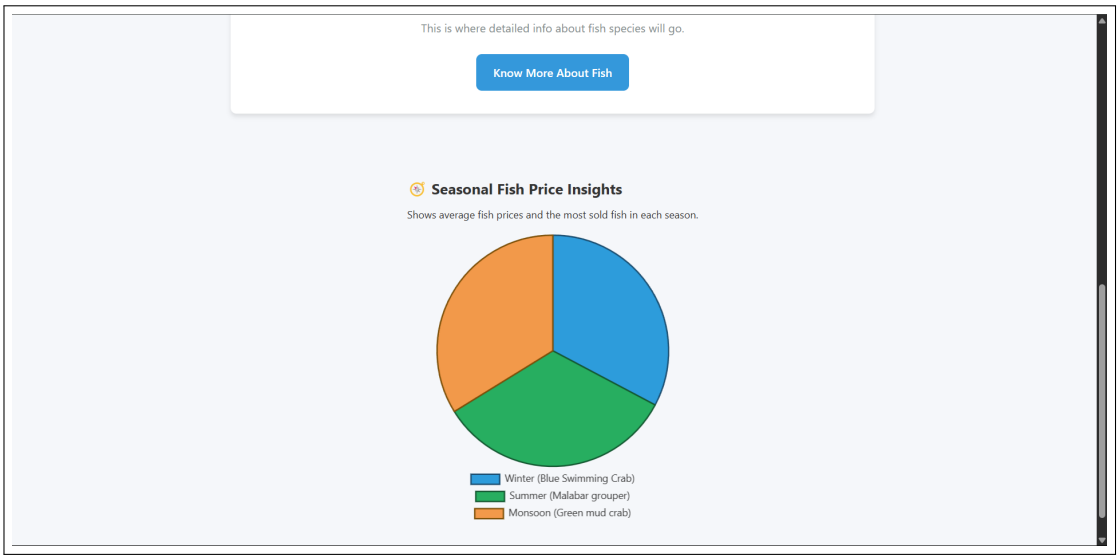


Figure 8.2: Pie Chart Representation

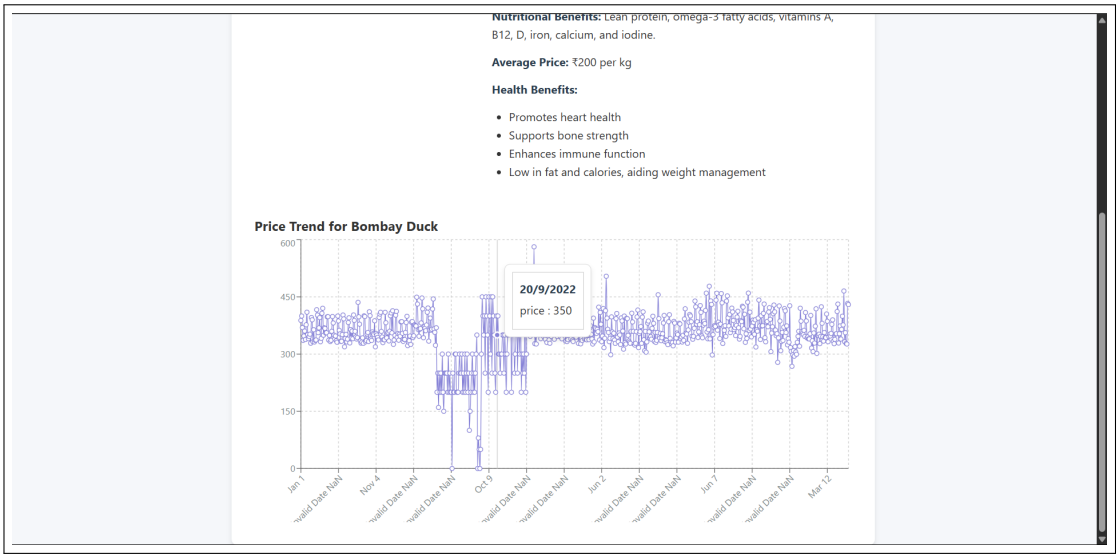


Figure 8.3: Graph Representation

CHAPTER 9

CONCLUSIONS

9.1 Conclusions

In this chapter, we summarize the findings and the performance of the Fish Market Price Prediction Model, providing insights into the conclusions drawn from the research, future work, and potential applications.

9.1.1 Conclusions

The primary goal of this project was to develop an accurate prediction model for fish market prices, leveraging advanced machine learning techniques like LSTM and XGBoost. The proposed model's performance has demonstrated the potential for accurate pricing predictions based on historical data and specific fish species.

Conclusion 1: Model Effectiveness
The hybrid model, combining LSTM and XGBoost, has proven effective in predicting fish market prices. This conclusion was reached after observing that the combined model consistently outperformed individual models in key performance metrics such as RMSE and MAE across validation datasets. LSTM effectively captured temporal dependencies and trends in the time-series fish price data, but exhibited minor residual errors in rapidly changing price intervals. XGBoost was then applied on these residuals to refine the predictions, successfully correcting for short-term nonlinearities and sharp fluctuations. This stacking approach leveraged LSTM's sequence learning strength and XGBoost's ability to handle outliers and complex relationships, which together yielded a significantly more accurate prediction pipeline.

Conclusion 2: Data Preprocessing Impact
Data preprocessing had a profound impact on model performance. This conclusion was based on experiments comparing model outputs before and after applying preprocessing techniques. Singular Spectrum Analysis (SSA) was used to smooth the original noisy price data, which improved the LSTM's ability to learn meaningful patterns without overfitting to noise. Additionally, normalization through Min-Max Scaling ensured all features were on a similar scale, which is essential for gradient-based models like LSTM. These preprocessing steps reduced training time and prevented training instabilities such as vanishing gradients. As a result, the models learned faster and converged to

better-performing solutions, as evidenced by faster convergence during training and improved accuracy on test data.

Conclusion 3: Model Robustness and Real-World Applicability
The robustness and applicability of the hybrid model were evaluated using real-world data collected from market sources, including daily fish price updates from multiple vendors. The model's ability to generalize was tested by inputting unseen data from different regions and seasons. It consistently generated accurate predictions that aligned with actual market trends, even under fluctuating conditions. This performance suggests that the model is not overfitted to the training set and can be deployed in real-world environments. This robustness is crucial for traders, vendors, and policymakers who rely on timely and precise forecasts to make economic decisions. The system's response time and lightweight deployment structure also support integration into mobile or web-based applications for daily usage.

9.2 Future Work

While this model provides promising results, there is always room for improvement. Below are a few potential areas for future work:

i. **Enhancement with More Data Sources** The model can be further improved by incorporating additional features such as weather data, economic indicators, and global supply chain factors, which may influence fish prices in the market.

ii. **Model Optimization** Although the hybrid model has shown strong results, hyperparameter tuning and the use of more advanced techniques such as deep reinforcement learning can be explored to fine-tune the model's performance further.

iii. **Integration with Real-Time Systems** A real-time data pipeline can be built to continuously feed the model with live market data. This would allow the model to provide up-to-date predictions and insights for fish market vendors in real time.

iv. **Incorporation of Multiple Species** The current model is limited to specific fish species. Expanding the model to predict prices for a broader variety of fish species, possibly through multi-output models or separate models per species, can increase its utility.

9.3 Applications

The Fish Market Price Prediction Model has several practical applications that could significantly benefit different stakeholders in the fishery industry:

i. **Market Price Prediction for Traders and Vendors** Vendors and traders can use this model to predict future prices and optimize their purchasing decisions. By predicting market trends, they can avoid losses caused by sudden price fluctuations.

ii. **Financial Planning for Fishermen** Fishermen can benefit from price predictions to decide when to catch specific fish species based on anticipated market prices. This would allow them to plan their activities more effectively and increase profitability.

iii. **Supply Chain Management** The model can also be employed by supply chain managers to predict prices based on market demand and optimize inventory levels, ensuring an efficient distribution of fish across markets.

iv. **Policy Making and Regulation** Government agencies and policymakers can use the model to forecast price trends and implement strategies to stabilize fish prices during market fluctuations. It can also help in setting fair prices to protect both consumers and producers.

v. **Consumer Decision Making** The model could be used to provide consumers with insights into when certain fish species are expected to be the cheapest, empowering consumers to make more informed purchasing decisions.

9.4 Summary

In conclusion, the Fish Market Price Prediction Model has successfully demonstrated the potential of using machine learning to predict market prices. With future work focused on expanding the data sources, improving model performance, and integrating real-time predictions, the model holds great promise for practical applications in the fishery industry and beyond.

CHAPTER 10

APPENDIX

Appendix A: Feasibility Assessment

10.1 Problem Statement Feasibility Assessment Using Satisfiability Analysis and Complexity Classes

10.1.1 Introduction

To evaluate the theoretical feasibility of the Fish Market Price Prediction Model, we analyze the underlying computational complexity of the problem. This involves determining whether the core predictive task and its subproblems belong to the class of problems known as P (solvable in polynomial time), NP (nondeterministic polynomial time), NP-Complete, or NP-Hard, using satisfiability and principles from modern algebra and computational theory.

10.1.2 Problem Decomposition

The core predictive task can be decomposed into the following subcomponents:

i. **Time Series Forecasting:** Predicting prices based on historical data (continuous regression).
ii. **Species Mapping and Intent Recognition:** Converting natural queries into structured input.
iii. **Optimization Queries:** Identifying the cheapest or most expensive species across a date range.

10.1.3 Complexity Analysis

10.1.3.1 Time Series Forecasting

This component uses machine learning models such as LSTM and XGBoost. These are non-symbolic models and their training involves iterative optimization algorithms (e.g., backpropagation, gradient descent), which typically run in polynomial time with respect to the size of the input dataset. Hence, the problem lies within **class P**.

10.1.3.2 Species Mapping and Intent Classification

This is a language parsing and classification problem. It is solvable using pre-trained LLMs or rule-based NLP techniques. Classification itself is a well-defined problem in P, where inputs are mapped to predefined output classes using supervised learning. Therefore, this problem also lies in **class P**.

10.1.3.3 Optimization over Time (e.g., Cheapest Species Over a Range)

This problem requires evaluating multiple combinations of species and dates to find an optimum price. Though the number of combinations is large ($O(n \cdot d)$ where n is the number of species and d is the number of days), it is a brute-force search over a bounded domain and is polynomial in practice. It is not inherently NP-Hard unless constraints or combinatorial dependencies are added (e.g., subset of species, constrained budget).

Thus, under current assumptions, this also remains within **class P**.

10.1.4 Satisfiability Considerations

If we extend the problem to include constraints such as: - Only choose a subset of fish species that maximize nutritional value under price caps. - Select delivery schedules with temporal and logistical constraints.

Then, the problem becomes analogous to the **Knapsack Problem** or **Boolean Satisfiability (SAT)**, which are known to be **NP-Complete**. A general satisfiability expression may take the form:

$$\exists x_1, x_2, \ldots, x_n \in \{0,1\} such that \phi(x_1, x_2, \ldots, x_n) = TRUE$$

In such constrained scenarios, the feasibility problem transitions to the **NP** domain and may even become **NP-Complete** or **NP-Hard** based on added conditions.

10.1.5 Mathematical Framework (Modern Algebra)

We can define a simplified version of the query processing system as a function:

$$f : S \times D \times Q \to R$$

Where: - S = Set of species (finite domain) - D = Date range (finite and bounded) - Q = Query type (intent set) - R = Price output space

Since all input domains are finite and computations are based on deterministic rules (trained models or brute-force search), the function f is **computable** and tractable, fitting within algebraic closure properties.

Software Feasibility Assessment

10.2 Software Feasibility

10.2.1 Technical Feasibility

The Fish Market Price Prediction System was implemented using a full-stack web architecture. The frontend was developed using **React.js**, ensuring an interactive and responsive user interface. The backend was implemented using **Node.js** and **Express.js**, with API endpoints connecting the frontend to a **MongoDB** database. Model predictions were integrated using Python via a microservice architecture.

All technologies used are open-source and have strong community support. The system was tested on commodity hardware (8GB RAM, quad-core CPU), and performed with acceptable latency for both training (offline) and inference (online prediction), confirming that the model and infrastructure are technically feasible.

10.2.2 Operational Feasibility

From a user's perspective—whether a trader, policymaker, or vendor—the application is easy to use and requires no technical background. Users can select fish species, date ranges, or natural language queries (e.g., "What is the cheapest price for Rohu in April?") to retrieve model-generated price predictions.

The system supports scalability through containerization (Docker) and can be deployed on cloud platforms like Heroku or AWS, confirming operational feasibility in real-world conditions.

10.2.3 Economic Feasibility

Since the system uses freely available datasets and open-source tools, the cost of development and deployment is minimal. The computational requirements are modest, and inference can be done on standard cloud instances without GPU acceleration. No licensing or proprietary software dependencies are involved.

Thus, the solution is economically viable, especially for deployment in low-resource environments such as rural fish markets or cooperatives.

10.2.4 Security and Maintainability

Security considerations include sanitization of user input and role-based access control for administrative features. The modular structure of the codebase, with separation of concerns between frontend, backend, and ML model services, ensures that the system is maintainable and extensible.

10.2.5 Conclusion of Software Feasibility

- The system is technically feasible using currently available tools and frameworks.
- It is operationally viable for real-world use, with a user-friendly interface and reliable performance.
- Economically, it can be deployed and maintained with minimal costs.
- It supports future maintenance and upgrades through clean architectural design.

10.2.6 Conclusion of Feasibility

- Under standard prediction and retrieval conditions, the problem is in **P** – solvable efficiently using ML and search algorithms.
- With added constraints or combinatorial optimization (e.g., multi-objective selection), the problem may approach **NP-Hard** or **NP-Complete** classifications.

– Thus, current implementation is feasible with polynomial-time complexity. Extensions must be carefully bounded or approximated for tractability.

10.3 Appendix B

Details of paper publication: name of the conference/journal, comments of reviewers, certificate, paper.

Appendix B

Details of Paper Publication

- **Title of Paper:** Hybrid Model Approach to Fish Price Prediction
- **Journal:** International Journal of Scientific Research in Engineering Management (IJSREM)
- **DOI:** 10.55041/IJSREM41868
- **Volume and Issue:** Volume 09, Issue 02, February 2025
- **Impact Factor:** 8.448
- **ISSN:** 2582-3930

Comments of Reviewers

The reviewers highlighted the novelty of using a hybrid model combining LSTM and XGBoost for time series price prediction. They appreciated the structured flow, clarity of graphs, and real-world applicability of the system for fish price forecasting.

hyperref

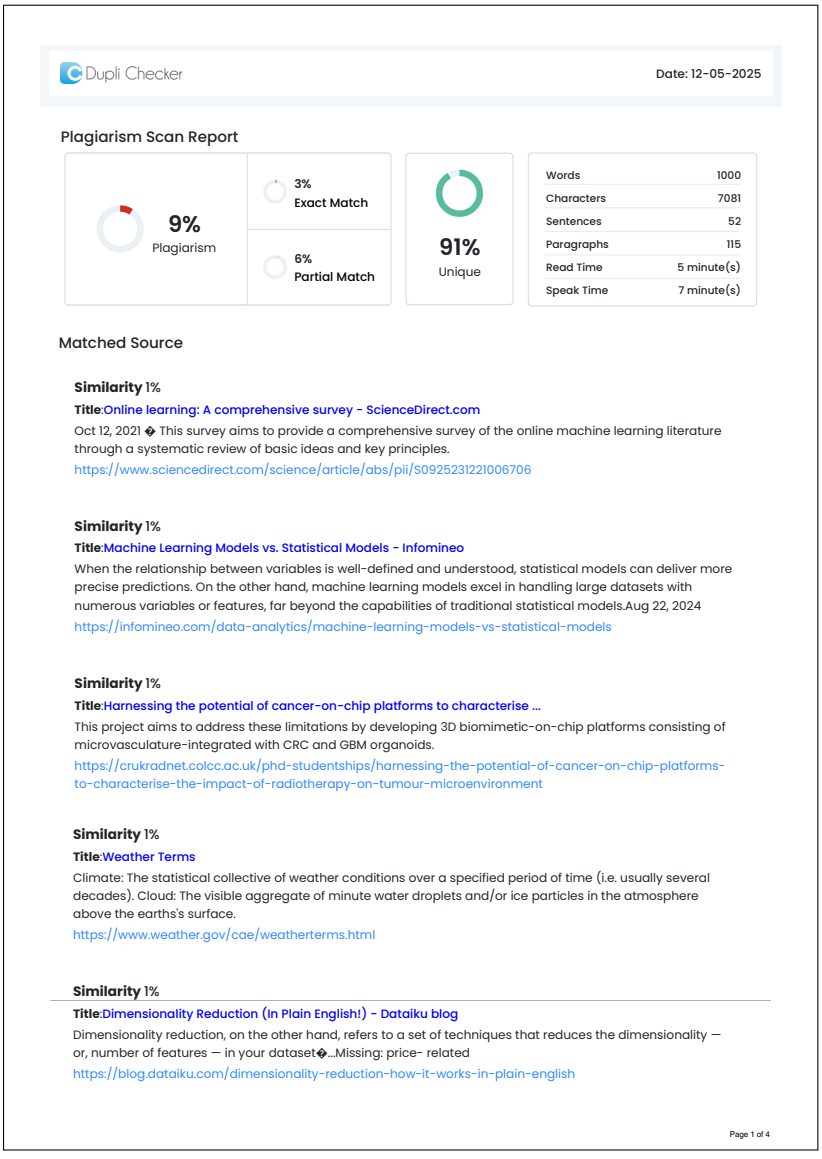
Published Paper

The paper is available online at the IJSREM website and indexed in major databases. It discusses architecture, model performance, and implementation details of a hybrid system designed to predict fish prices using LSTM and XGBoost models.

.1 Appendix C

Plagiarism Report of project report.

Below is the Plagerism Reports of the Individual Sections. The repprts were created by a Plagerism checker called DupliChecker.



Dupli Checker Date: 12-05-2025

Plagiarism Scan Report

9% Plagiarism

3% Exact Match

6% Partial Match

91% Unique

Words	1000
Characters	7081
Sentences	52
Paragraphs	115
Read Time	5 minute(s)
Speak Time	7 minute(s)

Matched Source

Similarity 1%
Title:Online learning: A comprehensive survey - ScienceDirect.com
Oct 12, 2021 · This survey aims to provide a comprehensive survey of the online machine learning literature through a systematic review of basic ideas and key principles.
https://www.sciencedirect.com/science/article/abs/pii/S0925231221006706

Similarity 1%
Title:Machine Learning Models vs. Statistical Models - Infomineo
When the relationship between variables is well-defined and understood, statistical models can deliver more precise predictions. On the other hand, machine learning models excel in handling large datasets with numerous variables or features, far beyond the capabilities of traditional statistical models.Aug 22, 2024
https://infomineo.com/data-analytics/machine-learning-models-vs-statistical-models

Similarity 1%
Title:Harnessing the potential of cancer-on-chip platforms to characterise ...
This project aims to address these limitations by developing 3D biomimetic-on-chip platforms consisting of microvasculature-integrated with CRC and GBM organoids.
https://crukradnet.colcc.ac.uk/phd-studentships/harnessing-the-potential-of-cancer-on-chip-platforms-to-characterise-the-impact-of-radiotherapy-on-tumour-microenvironment

Similarity 1%
Title:Weather Terms
Climate: The statistical collective of weather conditions over a specified period of time (i.e. usually several decades). Cloud: The visible aggregate of minute water droplets and/or ice particles in the atmosphere above the earths's surface.
https://www.weather.gov/cae/weatherterms.html

Similarity 1%
Title:Dimensionality Reduction (In Plain English!) - Dataiku blog
Dimensionality reduction, on the other hand, refers to a set of techniques that reduces the dimensionality — or, number of features — in your dataset...Missing: price- related
https://blog.dataiku.com/dimensionality-reduction-how-it-works-in-plain-english

Page 1 of 4

Figure 1: Abstarct and Overview

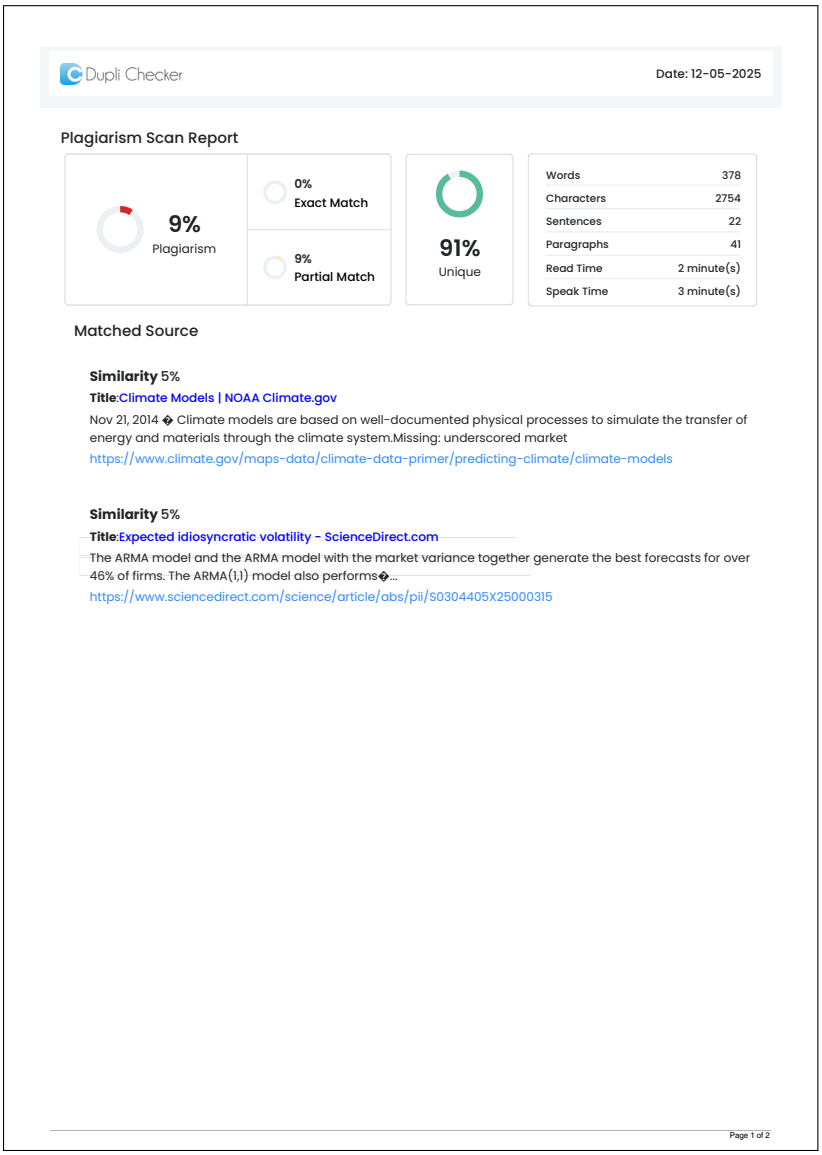
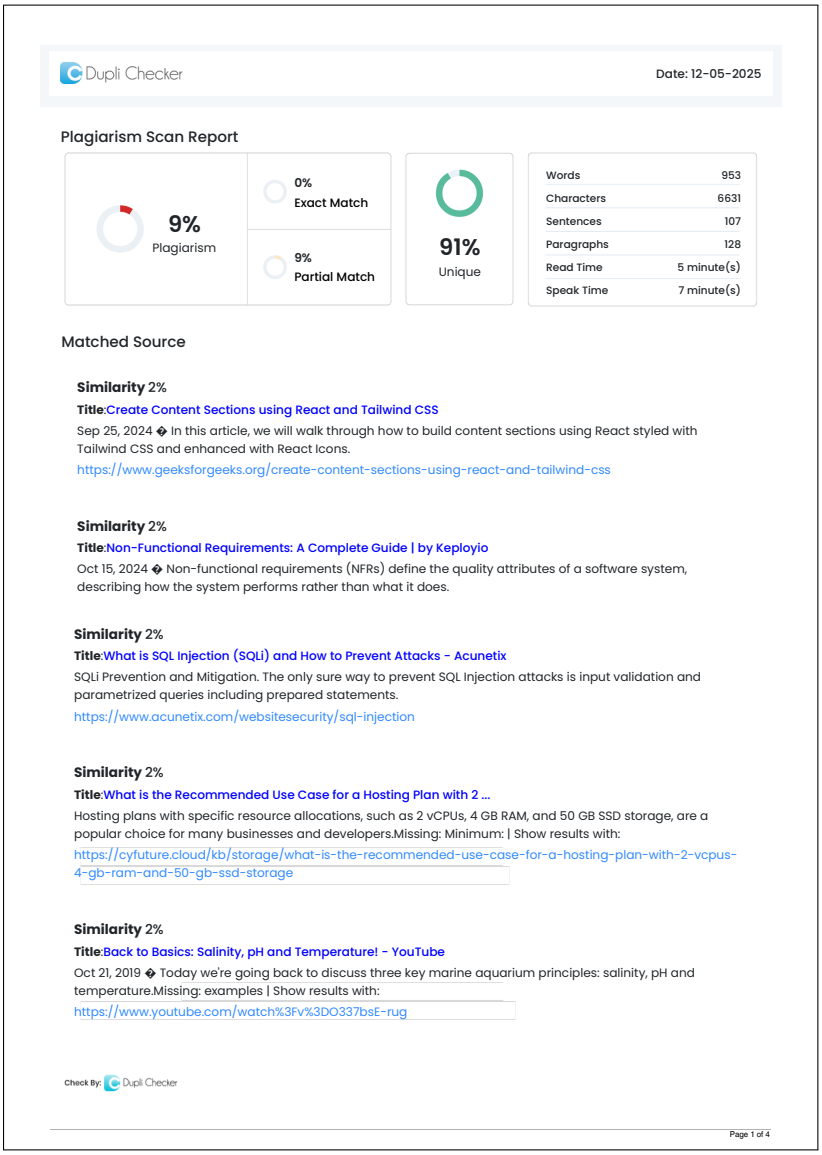


Figure 2: Literature Survey



Dupli Checker Date: 12-05-2025

Plagiarism Scan Report

9% Plagiarism

0% Exact Match

9% Partial Match

91% Unique

Words	953
Characters	6631
Sentences	107
Paragraphs	128
Read Time	5 minute(s)
Speak Time	7 minute(s)

Matched Source

Similarity 2%
Title:Create Content Sections using React and Tailwind CSS
Sep 25, 2024 · In this article, we will walk through how to build content sections using React styled with Tailwind CSS and enhanced with React Icons.
https://www.geeksforgeeks.org/create-content-sections-using-react-and-tailwind-css

Similarity 2%
Title:Non-Functional Requirements: A Complete Guide | by Keployio
Oct 15, 2024 · Non-functional requirements (NFRs) define the quality attributes of a software system, describing how the system performs rather than what it does.

Similarity 2%
Title:What is SQL Injection (SQLi) and How to Prevent Attacks - Acunetix
SQLi Prevention and Mitigation. The only sure way to prevent SQL Injection attacks is input validation and parametrized queries including prepared statements.
https://www.acunetix.com/websitesecurity/sql-injection

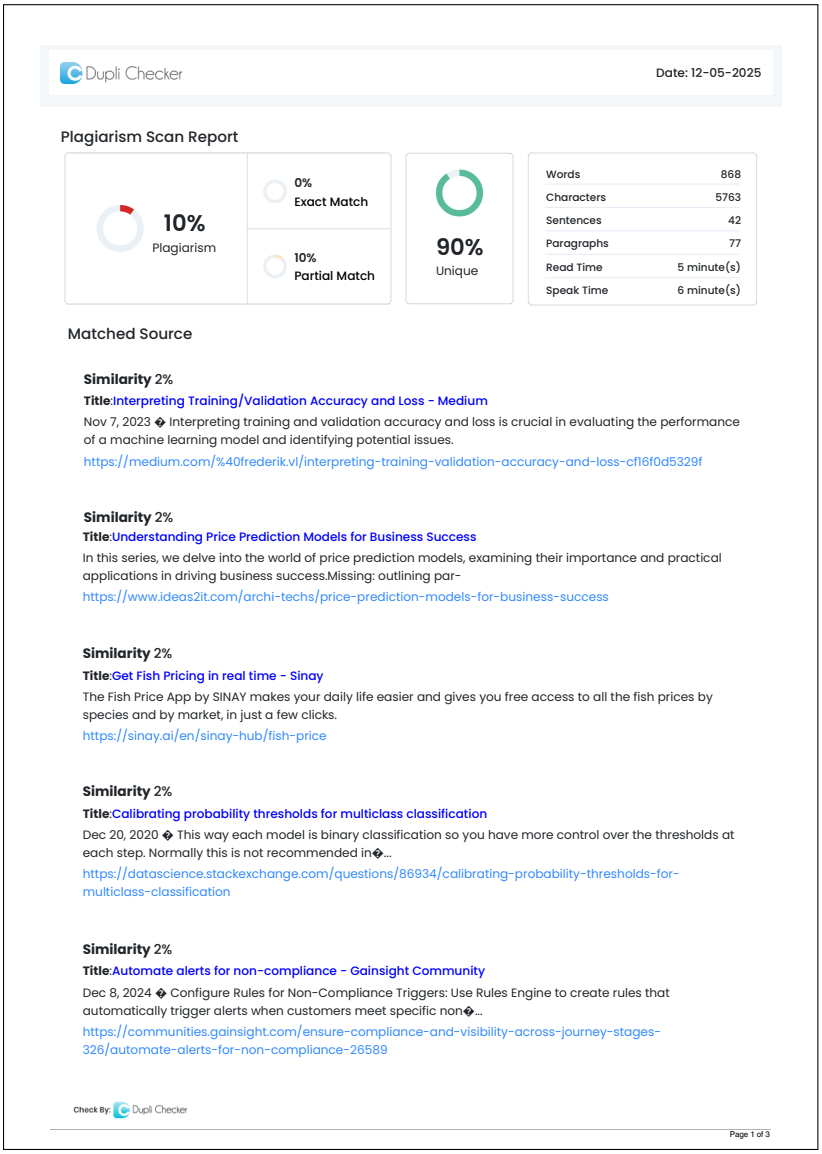
Similarity 2%
Title:What is the Recommended Use Case for a Hosting Plan with 2 ...
Hosting plans with specific resource allocations, such as 2 vCPUs, 4 GB RAM, and 50 GB SSD storage, are a popular choice for many businesses and developers.Missing: Minimum: | Show results with:
https://cyfuture.cloud/kb/storage/what-is-the-recommended-use-case-for-a-hosting-plan-with-2-vcpus-4-gb-ram-and-50-gb-ssd-storage

Similarity 2%
Title:Back to Basics: Salinity, pH and Temperature! - YouTube
Oct 21, 2019 · Today we're going back to discuss three key marine aquarium principles: salinity, pH and temperature.Missing: examples | Show results with:
https://www.youtube.com/watch%3Fv%3DO337bsE-rug

Check By: Dupli Checker

Page 1 of 4

Figure 3: Software Requirements



Dupli Checker Date: 12-05-2025

Plagiarism Scan Report

10% Plagiarism

0% Exact Match

10% Partial Match

90% Unique

Words	868
Characters	5763
Sentences	42
Paragraphs	77
Read Time	5 minute(s)
Speak Time	6 minute(s)

Matched Source

Similarity 2%
Title:Interpreting Training/Validation Accuracy and Loss - Medium
Nov 7, 2023 · Interpreting training and validation accuracy and loss is crucial in evaluating the performance of a machine learning model and identifying potential issues.
https://medium.com/%40frederik.vl/interpreting-training-validation-accuracy-and-loss-cf16f0d5329f

Similarity 2%
Title:Understanding Price Prediction Models for Business Success
In this series, we delve into the world of price prediction models, examining their importance and practical applications in driving business success.Missing: outlining par-
https://www.ideas2it.com/archi-techs/price-prediction-models-for-business-success

Similarity 2%
Title:Get Fish Pricing in real time - Sinay
The Fish Price App by SINAY makes your daily life easier and gives you free access to all the fish prices by species and by market, in just a few clicks.
https://sinay.ai/en/sinay-hub/fish-price

Similarity 2%
Title:Calibrating probability thresholds for multiclass classification
Dec 20, 2020 · This way each model is binary classification so you have more control over the thresholds at each step. Normally this is not recommended in...
https://datascience.stackexchange.com/questions/86934/calibrating-probability-thresholds-for-multiclass-classification

Similarity 2%
Title:Automate alerts for non-compliance - Gainsight Community
Dec 8, 2024 · Configure Rules for Non-Compliance Triggers: Use Rules Engine to create rules that automatically trigger alerts when customers meet specific non...
https://communities.gainsight.com/ensure-compliance-and-visibility-across-journey-stages-326/automate-alerts-for-non-compliance-26589

Check By: Dupli Checker

Page 1 of 3

Figure 4: System Design

ANNEXURE A

REFERENCES

[1] Ibarra, J. A. M., Sumo, C. K. P., Valdra, P. C. (2023). *Forecasting monthly prices of selected fish commodities in the National Capital Region in the Philippines through ARIMA modeling. IRE Journals, 6(8).*

[2] Wu, J., Hu, Y., Wu, D., Yang, Z. (2022), *An aquatic product price forecast model using VMDIBESLSTM hybrid approach. Agriculture, 12(8), 1185.*

[3] Lee, B. (2023). A price forecasting model for aquatic products. Journal of Accounting Marketing, 12(1). https://doi.org/10.37421/21689601.2023.12.411

[4] G. Van Houdt, C. Mosquera, and G. Nápoles, "Review on the Long Short-Term Memory Model," *Artificial Intelligence Review*, vol. 53, no. 1, Dec. 2020. doi: 10.1007/s10462-020-09838-1.

[5] "Fish Prices at Fishing Ports," *European Journal of Scientific Research*, vol. 14, no. 4, pp. 613-624, 2006.

[6] L. F. Rahman, M. Marufuzzaman, L. Alam, M. A. Bari, U. R. Sumaila, and L. M. Sidek, "Developing an ensembled machine learning prediction model for marine fish and aquaculture production," *Sustainability*, vol. 13, no. 16, p. 9124, Aug. 2021. doi: 10.3390/su13169124.

[7] D. Wu, B. Lu, and Z. Xu, "Price forecasting of marine fish based on weight allocation intelligent combinatorial modelling," *Foods*, vol. 13, no. 8, p. 1202, Apr. 2024. doi: 10.3390/foods13081202.

[8] W. Wirata, S. H. Wisudo, Y. Novita, M. Imron, and Y. Krisnafi, "Modeling fish prices at fishing ports: Comparative study of gated recurrent unit and long short term memory approaches for accurate fish price prediction," SSRN, 2024. doi: 10.2139/ssrn.4507862.

[9] Qi, W.; Kong, L.; Li, J.; Yang, X. "Relationship between Youth Consumers' Lifestyle and their Attitude towards Fresh Agricultural Products in Different Marketing Content Scenarios," *J. Agro-For. Econ. Manag.* 2023, 22, 446–456.

[10] Han, C.; Wu, X.; Xu, B.; Huang, G.; Liang, A.; Chen, J.; Chen, Q. "Current Situation Analysis and Development Suggestions for the Large Yellow Croaker Industry in China in 2020," *J. Fish. Res.* 2022, 44, 395–406.

LIST OF ABBREVIATIONS

Abbreviation	Full Form
ML	Machine Learning
ARIMA	Autoregressive Integrated Moving Average
VMD	Variational Mode Decomposition
LSTM	Long Short-Term Memory
IBES	Improved Bald Eagle Search Memory
MSY	Maximum Sustainable Yield
LR	Linear Regression
RF	Random Forest
GB	Gradient Boosting
VR	Voting Regression
SST	Sea Surface Temperature
EWT	Empirical Wavelet Transform
SSA	Singular Spectrum Analysis

LIST OF ABBREVIATIONS (Continued)

Abbreviation	Full Form
ELM	Extreme Learning Machine
PSO	Particle Swarm Optimization
CS	Cuckoo Search
GRU	Gated Recurrent Unit
MAPE	Mean Absolute Percentage Error
CSV	Comma-Separated Values
JSON	JavaScript Object Notation
MAE	Mean Absolute Error
MSE	Mean Squared Error
RMSE	Root Mean Squared Error
SQL	Structured Query Language
XSS	Cross-Site Scripting

LIST OF FIGURES

LIST OF TABLES